Committed to Education:

School Fees, Vouchers, and the Timing of Cash Transfers

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October 9, 2026

Schooling costs remain a barrier to primary-school completion

- Approximately **1 in 3 children** in Sub-Saharan Africa do not complete primary school by age 15 (UNESCO GEM 2022)
- Schooling costs are widely viewed as a major barrier to participation (Deininger 2003; Schultz 2004; Evans et al. 2009; Baird et al. 2011; Duflo et al. 2015)
- In Uganda, among children who leave school, **60%** cite schooling costs as the main reason (Uganda Bureau of Statistics 2025)

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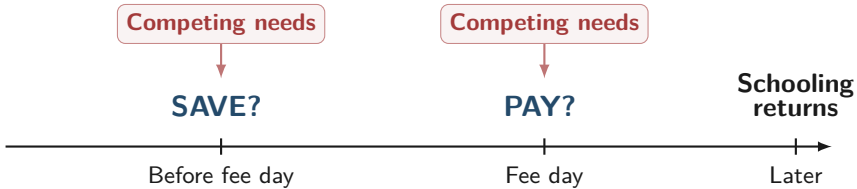
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→ **Non-payment may reflect competing priorities:** households may choose other current needs over paying school fees.

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⇒ **This study:** choose the timing/form of a cash transfer as a commitment device to restrict future saving or spending

This study

Setting: 2,000 primary-school children and their parents in Eastern Uganda

1. **Are parents willing to pay a premium to restrict the timing/form of a transfer?**
2. **How do these transfers affect schooling outcomes?**
3. **Does parental demand identify who responds most to these transfers?**

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3. **Does parental demand identify who responds most to these transfers?**
   - Link beliefs and demand to realized treatment effects
   - Re-elicit choices for a future Voucher vs. Fee-Day Cash

Contributions to the literature

1. **Schooling costs and participation** Schultz 2004; Evans et al. 2009; Baird et al. 2011; Barrera-Osorio et al. 2011; Bursztyn & Coffman 2012; Benhassine et al. 2015; Duflo et al. 2015
2. **Self-control and commitment** Ashraf et al. 2006; Bryan et al. 2010; Giné et al. 2010; Kaur et al. 2015; Laibson 2015; Casaburi & Macchiavello 2019; Schilbach 2019; Brune et al. 2021; Allcott et al. 2022
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   - Link WTP for an intervention to its randomized effects, and show how incorrect beliefs can explain why the two diverge

Roadmap

1. Experimental Design
2. Conceptual Framework: Why pay for commitment?
3. Results
   - 3.1 Are parents willing to pay for commitment?
   - 3.2 How do these transfers affect schooling outcomes?
   - 3.3 Does parental demand identify who responds most to these transfers?
4. Conclusion

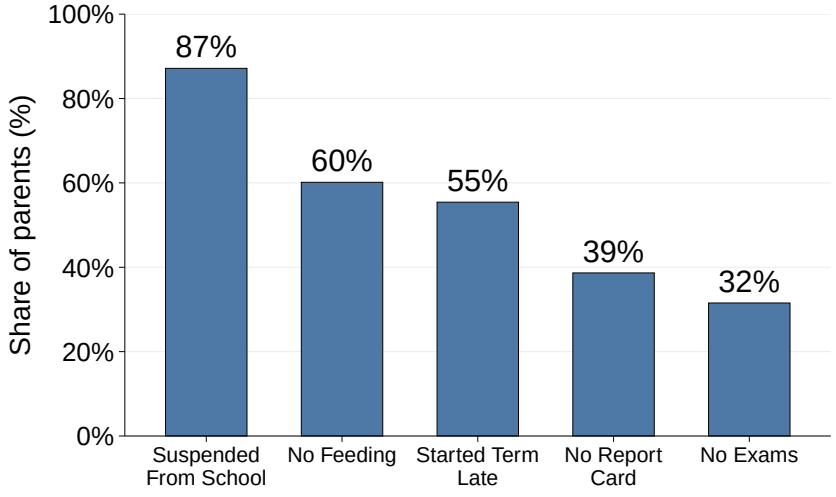
Experimental Context: Jinja District, Uganda



- 80% of students attend government (public) primary schools (UNESCO, 2017)
- Despite Universal Primary Education, households pay **term-by-term school fees** ("parent contributions")
  - Fees **due at start of each term**, but late/no payment is common
- **This study:** 2,000 pupils (57% female) and primary guardians (79% female, $\sim$ 50% mother) across 30 government schools
- Partner: Uganda Ministry of Education and Sports

sum stats fees by grade

In the last 12 months, did any of the following happen to [CHILD NAME] because school fees were paid late or not fully paid?



Experimental Design: Randomized transfers

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Arm	Transfer	Timing	Use of funds
Early Cash	35k UGX cash	∼6 weeks before term	Fungible
Fee-Day Cash	35k UGX cash	When term begins	Fungible
Voucher	35k UGX	When term begins	Paid directly to school
Control	—	—	No transfer

- 35k UGX ($26 USD PPP) ≈ 85% of average term fees ≈ 8% of median monthly HH consumption
- School admin blinded to cash vs. control assignment; design and primary outcomes preregistered (#AEARCTR-0016626)

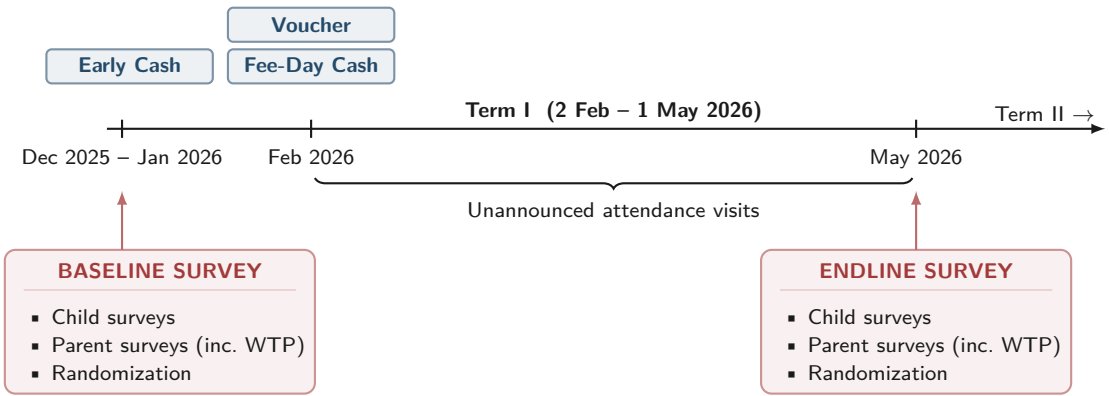
balance

Experimental Design: Randomized transfers

	Arm	Transfer	Timing	Use of funds	
Timing	Early Cash	35k UGX cash	∼6 weeks before term	Fungible	Full commitment
	Fee-Day Cash	35k UGX cash	When term begins	Fungible	
Earmarking	Voucher	35k UGX	When term begins	Paid directly to school	
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Experimental Design: Timeline



Experimental Design: Measuring demand for commitment

Before randomization, parents make two sets of incentive-compatible choices:

WTP for earmarking

A: 35k Voucher
B: X Fee-Day Cash

Vary X across choices to locate switch point X^*

$$\text{WTP}_{\text{earmark}} = X^* - 35{,}000$$

WTP for delayed cash

A: 35k Early Cash
B: Y Fee-Day Cash

Vary Y across choices to locate switch point Y^*

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elicitation details BDM

Experimental Design: Measuring demand for commitment

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Vary Y across choices to locate switch point Y^*

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- Incentive-compatible: 2% receive a transfer according to their choices
- Transaction costs equalized:
  1. Voucher requires a separate school visit
  2. Cash includes 1k UGX for mobile-money fees
- Order of elicitations randomized

elicitation details BDM

Why pay for commitment?

Open-ended interviews point to present focus

Parents describe protecting the money **from their own future selves**:

- *"I wouldn't want to be tempted to use the money to buy nice food like fish instead of paying it for school."*
- *"It is very easy to eat that money if you get it on [your] phone and you remain with [the] burden of paying fees at school."*
- *"Because if the money is paid directly to school its safe there than being with me whereby I can use it on other things."*

Competing mechanisms, including intra-household disagreement, kin taxes, theft, inflation are rarely brought up. $\rightarrow$ We test whether WTP varies with these mechanisms.

$\Rightarrow$ β–δ **framework to organize choices** (Laibson 1997; O'Donoghue & Rabin 1999)

Quasi-hyperbolic discounting model, tailored to the experiment

Quasi-hyperbolic preferences:

$$U_t = u(c_t) + \beta \sum_{k \geq 1} \delta^k u(c_{t+k}), \qquad \beta \leq 1$$

$\beta < 1$ captures present focus; $\beta = 1$ is time consistent benchmark

Model features:

- Linear utility, $u(c) = c$
- No exponential discounting, $\delta = 1$
- Uncertain income on fee day

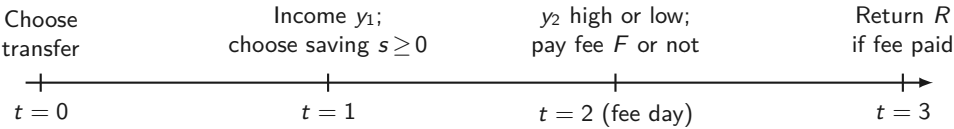
Two present-focus parameters:

- β: *true* present focus – governs actual behavior
- $\tilde{\beta}$: *perceived* present focus – governs anticipated behavior and WTP

$$\beta, \tilde{\beta} \in (0, 1], \qquad \tilde{\beta} = \beta \text{ sophisticated}, \quad \tilde{\beta} = 1 \text{ fully naive}$$

$\delta < 1$ concave utility borrowing

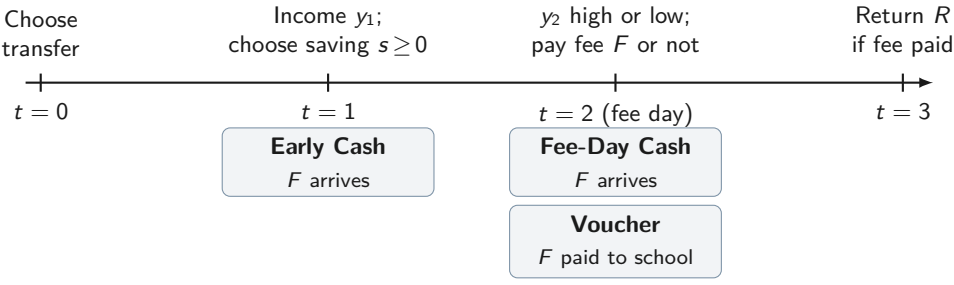
Model setup



Fee-day income is uncertain: $y_2 = y_H \geq F$ w/ prob. $1 - q$, else $y_L < F$ w/ prob. q.

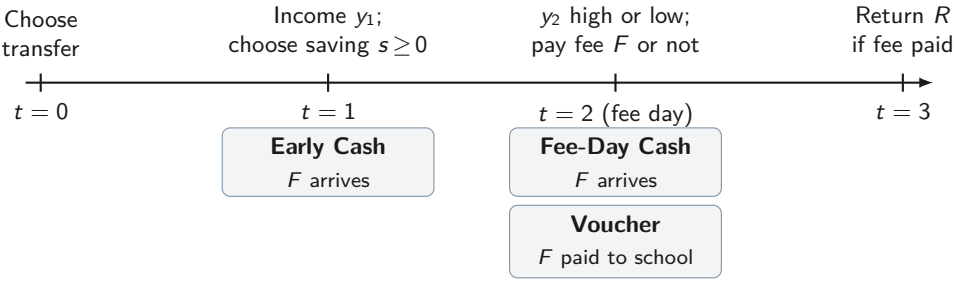
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Assumptions: $R > F$ (schooling worth it) ; $s \geq 0$ (save, can't borrow)

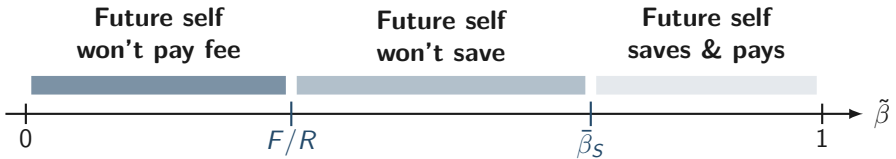
partial voucher

Who demands which restriction?

$$\mathbf{WTP}_d = (R - F)\,\Delta\tilde{\mu}_d \quad \text{net return} \times \text{perceived increase in prob. of paying the fee}$$

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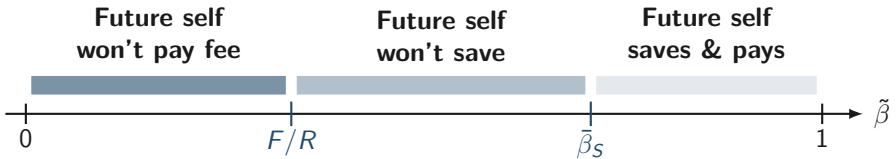
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$\tilde{\beta}R < F$:
pays for earmarking
$\mathbf{WTP}_{\text{earmark}} = R - F$

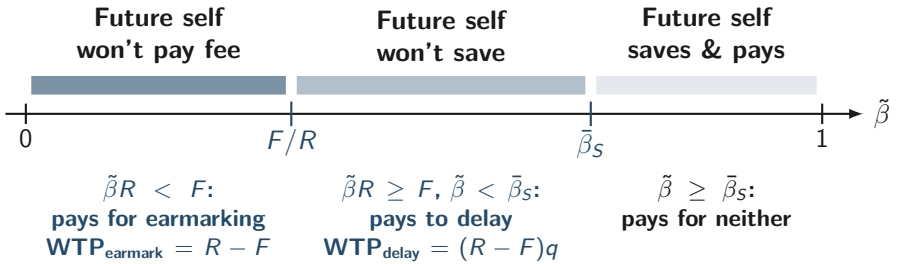
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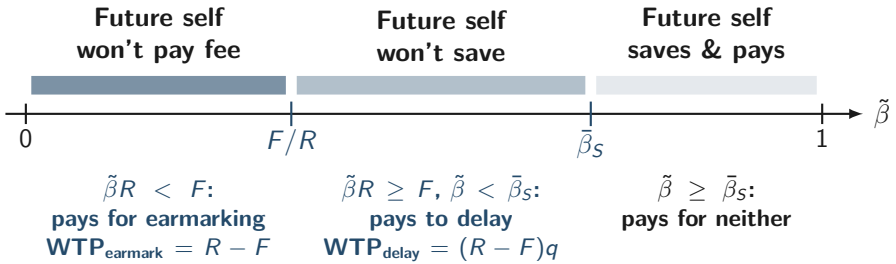
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$\rightarrow$ Positive WTP for either restriction implies anticipated present focus: $\tilde{\beta} < 1$

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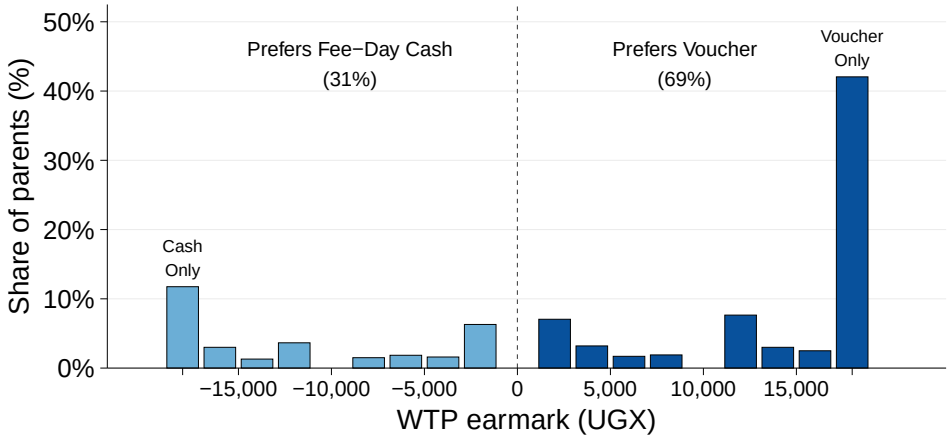


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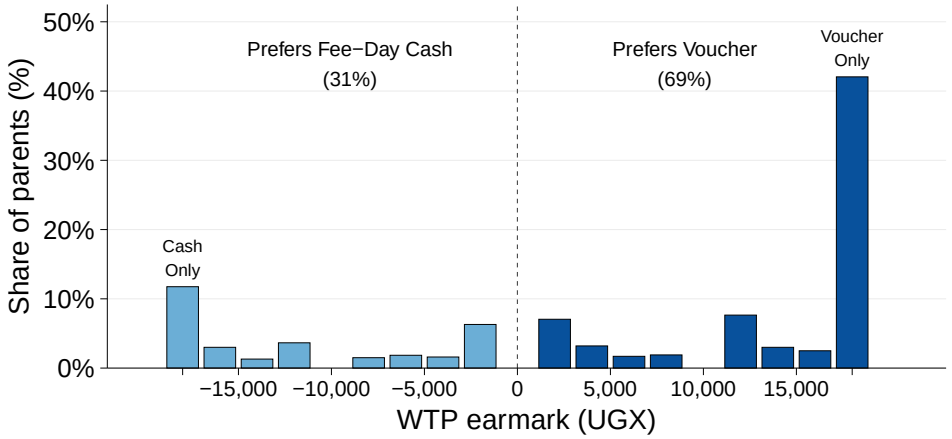
$\rightarrow$ Partial naïveté, $\beta < \tilde{\beta} < 1$: parents recognize the self-control problem but underestimate its severity (Heidhues & Kőszegi 2010)

Are parents willing to pay for commitment?

Parents prefer a Voucher to Fee-Day Cash

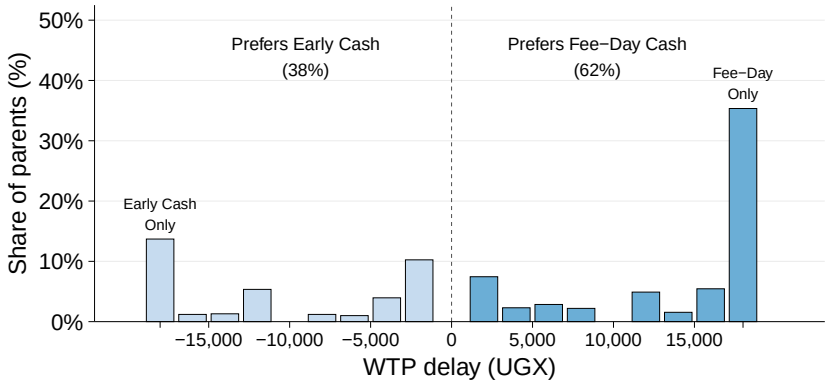


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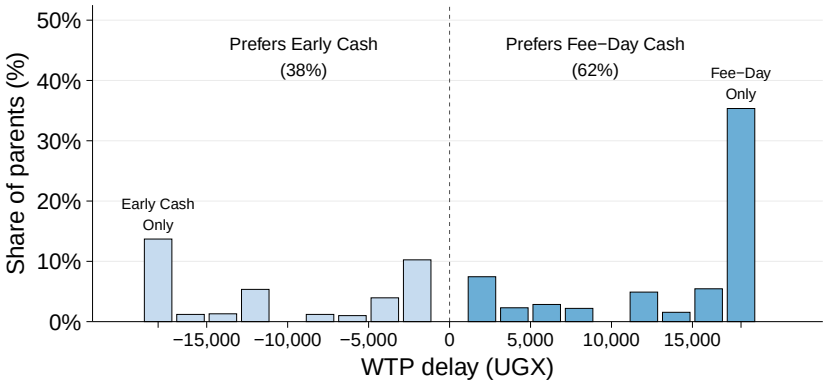
→ **69% are willing to pay for earmarking;** average WTP = 35% of the transfer value

Parents prefer Fee-Day Cash over Early Cash



joint distribution

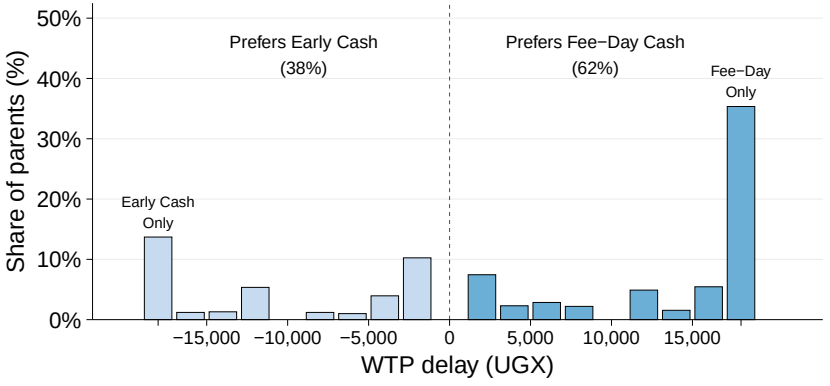
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⇒ **62% pay for delay;** average WTP = 24% of the transfer value

joint distribution

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⇒ **Model:** positive WTP for either restriction implies anticipated present focus, $\tilde{\beta} < 1$

joint distribution

Do parents want cash to move with the fee deadline?

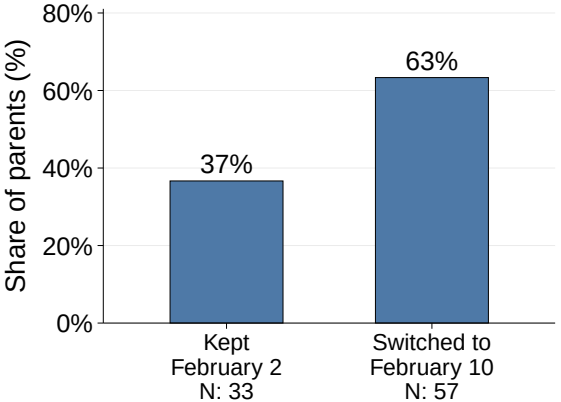
- After baseline, the government unexpectedly postponed Term 1 from **Feb 2 to Feb 10**.

Do parents want cash to move with the fee deadline?

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- We re-contacted 90 Fee-Day Cash parents and offered a **binding choice**:
  - Receive cash on Feb 2 as planned, or
  - Delay the same cash until Feb 10

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⇒ **63% choose to delay cash to the new fee date**

IV: does the date matter?

Measuring the perceived increase in the probability of paying

WTP $= (R - F)\Delta\tilde{\mu}$ = net return x perceived increase in prob. of paying the fee

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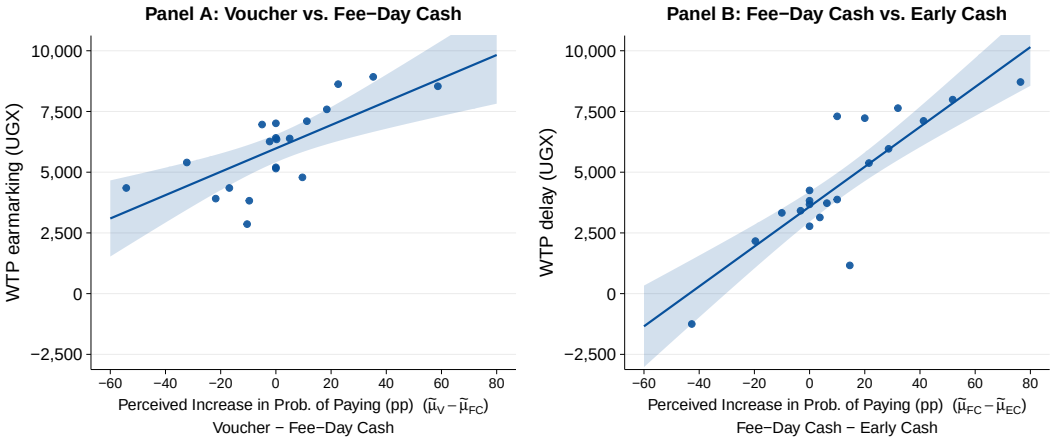
We directly elicit each parent's perceived probability of paying under each transfer.

Using a 0–100 ruler (0 = impossible, 100 = certain):

> *"Thinking about your current financial situation and everything your household is responsible for, from 0-100 how likely is it that you will fully clear [CHILD]'s Term 1 fees . . . "* $\rightarrow \tilde{\mu}_C$
>
> - . . . "if you receive a 35k voucher paid directly to the school?" $\rightarrow \tilde{\mu}_V$
> - . . . "if you receive 35k in cash today?" $\rightarrow \tilde{\mu}_{EC}$
> - . . . "if you receive 35k in cash on February 2?" $\rightarrow \tilde{\mu}_{FC}$

$\Rightarrow$ Perceived increase in probability of paying: $\tilde{\mu}_V - \tilde{\mu}_{FC}$ and $\tilde{\mu}_{FC} - \tilde{\mu}_{EC}$

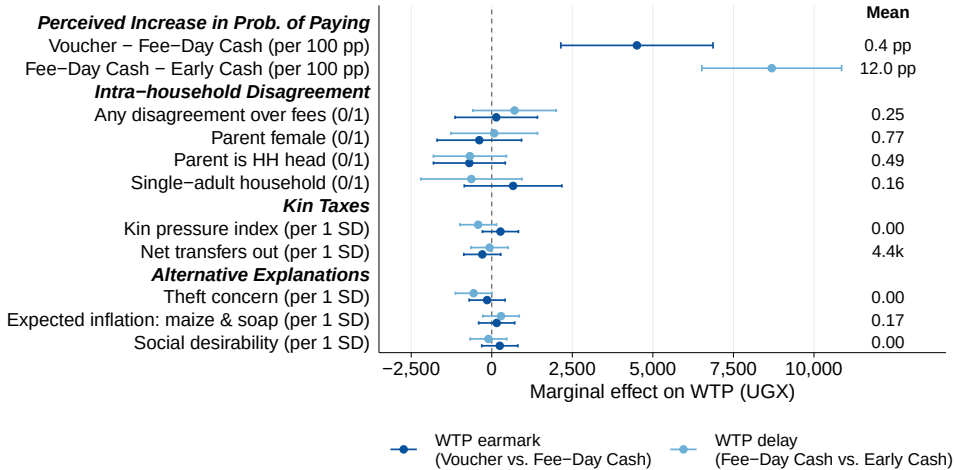
WTP rises with perceived increase in payment probability



Notes: Points are means within 20 equal-sized bins of the perceived increase in the probability of paying. ($N = 2{,}000$)

wealth gradient | net return to educ | controls

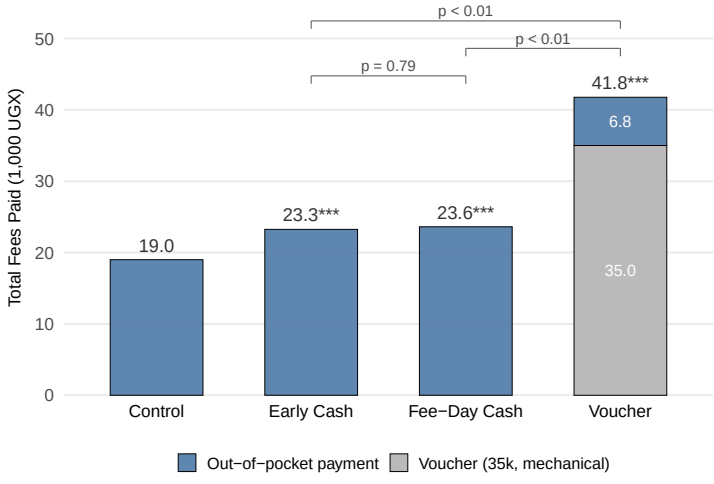
Demand is not explained by competing mechanisms



tobit specification | stated reasons | savings & borrowing

How do these transfers affect schooling outcomes?

While fee payments increase across all treatments, delayed cash is not an effective form of "commitment"



specification
share fees
fees cleared
Lee bounds
payment timing

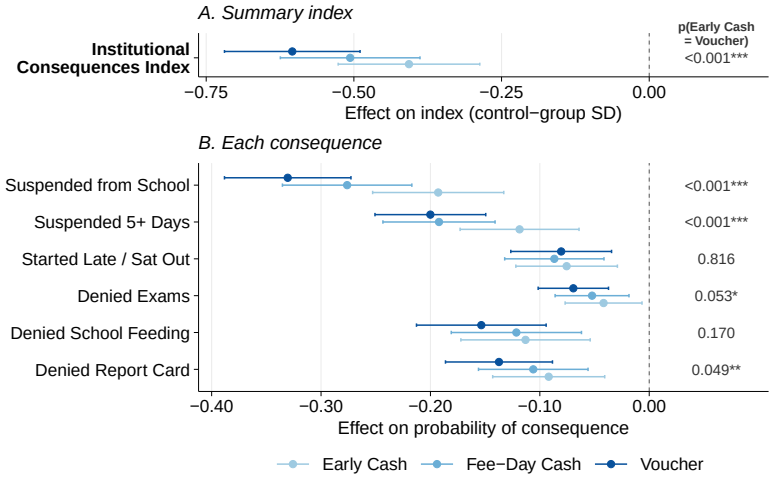
The voucher reduces within term dropout

	(1) Enrolled (Any School)	(2) Dropped Out (Did Not Finish)	(3) Transferred Schools
T1: Early Cash	0.002	-0.011	-0.028
	(0.008)	(0.013)	(0.019)
		[0.257]	[0.145]
T2: Fee-Day Cash	0.004	-0.015	-0.031*
	(0.007)	(0.013)	(0.019)
		[0.190]	[0.126]
T3: Voucher	0.009	-0.027**	-0.035*
	(0.007)	(0.012)	(0.019)
		[0.049]	[0.089]
Control Mean	0.984	0.051	0.119
Observations	1,960	1,960	1,960
T1 Effect (%)	0.2	-22.8	-23.4
T2 Effect (%)	0.4	-29.8	-26.1
T3 Effect (%)	0.9	-54.0	-29.4
p-value: T1 = T2	0.782	0.769	0.853
p-value: T2 = T3	0.384	0.248	0.819
p-value: T1 = T3	0.264	0.146	0.682

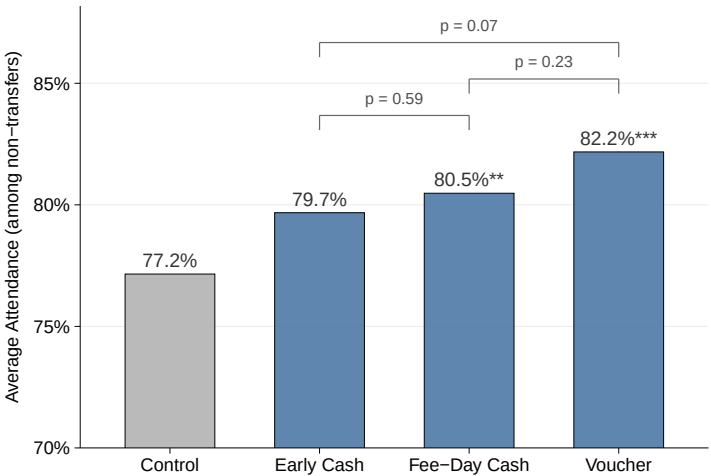
Notes: Parent endline survey. Strata FE, robust SEs in parentheses, sharpened *q*-values (Anderson 2008) in brackets.

literature comparison

All transfers reduce the incidence of children being sent home, missing exams, and other school consequences



The voucher produces modest attendance impacts



- Weekly unannounced attendance spot checks
- 2/3–3/4 of the attendance gain comes from better attendance among enrolled children (intensive margin)

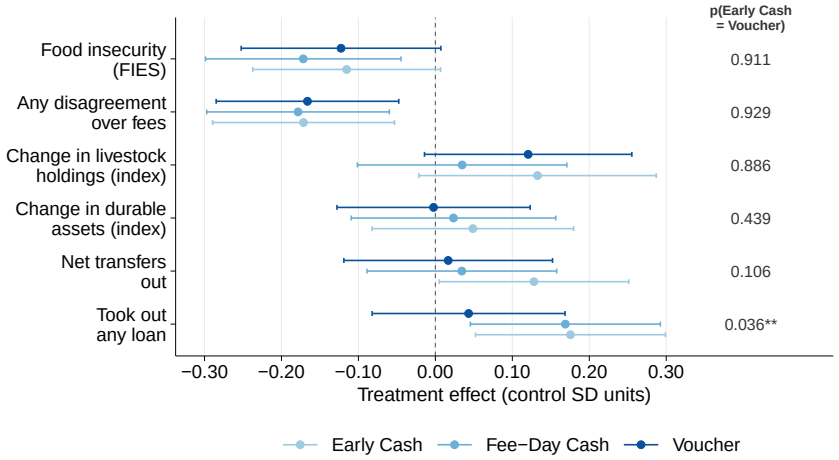
Lee bounds | decomposition | school spillovers | literature comparison

Additional schooling results and heterogeneities

- **Learning:** No detectable learning effects at short-run follow-up uwezo school exams
- **Socio-emotional:** No detectable effects on depressive symptoms or resilience socio-emotional
- **Child gender:** Same fee effects; dropout effect of vouchers concentrated among boys, attendance gains among girls (n.s.) child gender
- **Child grade:** Smaller Voucher effect on fees paid in Grade 6 than Grade 4 ($p < 0.01$), driven by crowd-out of own payments; no differences in dropout/attendance grade
- **Primary guardian:** Attendance gains larger when the caregiver is the mother (n.s.); no differences in fees or dropout mother
- **Household wealth:** No heterogeneity by household wealth wealth

Where does the voucher crowd out other HH spending?

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Transfers out

HH spillovers

Taking stock so far

1. **Are parents willing to pay a premium to restrict the timing/form of a transfer?**
   - **Yes:** 69% pay for earmarking, 62% for delayed cash
2. **How do these transfers affect schooling outcomes?**
   - **Voucher has the largest effects:** higher fee payment, only arm to reduce dropout
   - **Cash timing matters little:** Fee-Day Cash $\approx$ Early Cash, despite demand for delay

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$$\text{WTP}_{id} = (R_i - F_i)\,\Delta\tilde{\mu}_{id} \qquad \text{whereas} \qquad \text{Effect on fee clearing}_{id} = \Delta\mu_{id}$$

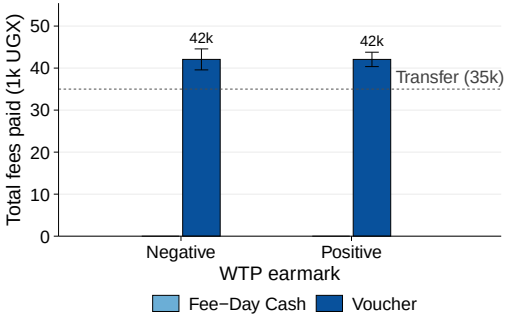
Does parental demand identify who responds most to these transfers?

Does WTP predict treatment effects on schooling outcomes?

$\rightarrow$ **If demand targets schooling gains,** the Voucher's effect should be *larger* among parents with $(+)$ WTP for earmarking

Does WTP predict treatment effects on schooling outcomes?

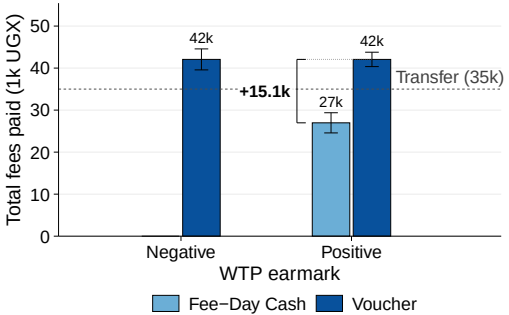
→ **If demand targets schooling gains,** the Voucher's effect should be *larger* among parents with (+) WTP for earmarking



all arms

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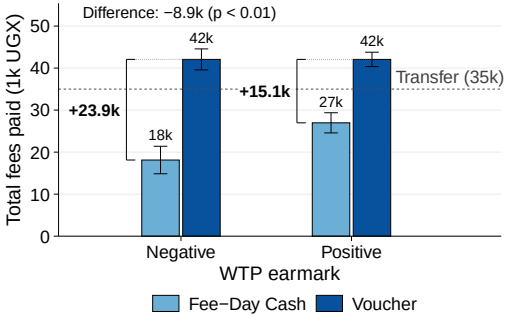
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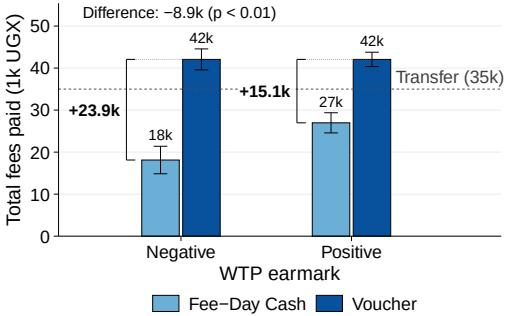
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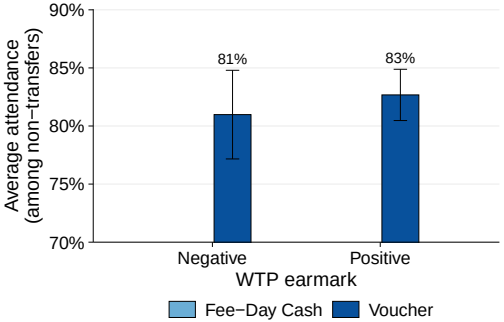


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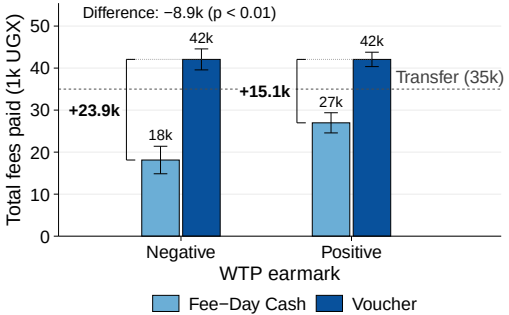


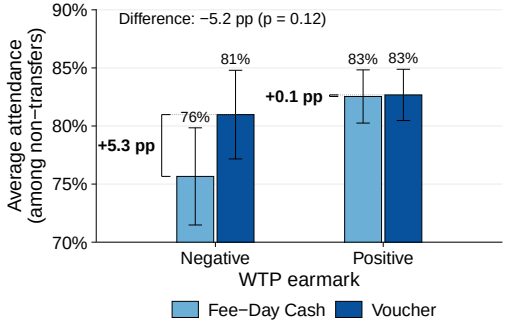


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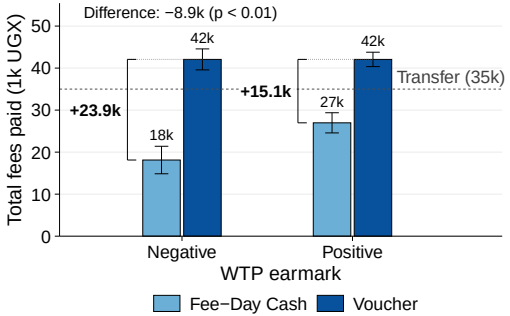


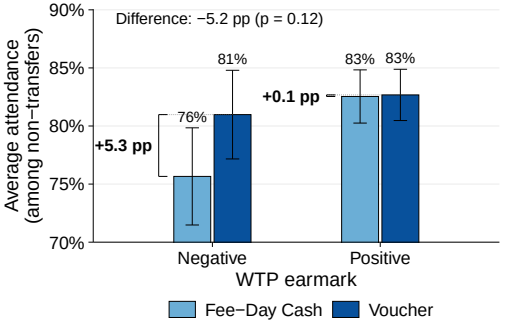


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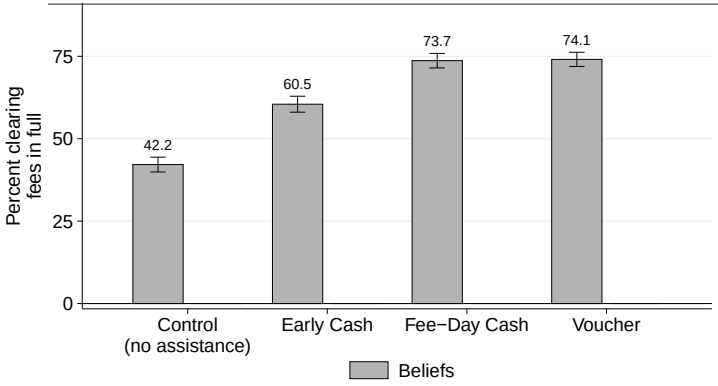


⇒ Demand is not a sufficient statistic for treatment effects (Walters 2018; Carrera et al. 2022)

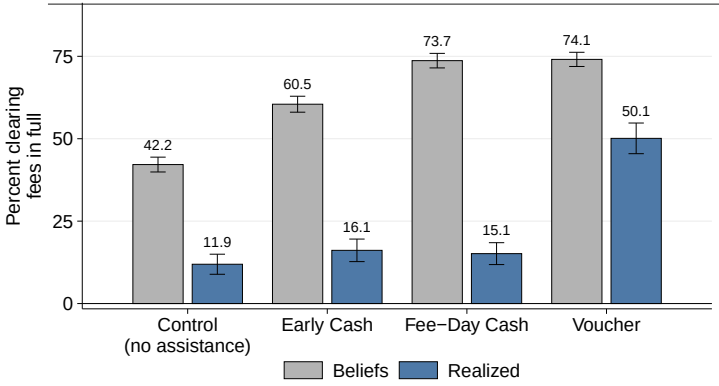
⇒ **Why?** $R - F$ differs, and/or parents mispredict what commitment does ($\Delta\tilde{\mu}_d \neq \Delta\mu_d$)

all arms

How accurate are parents' forecasts of fee clearing?



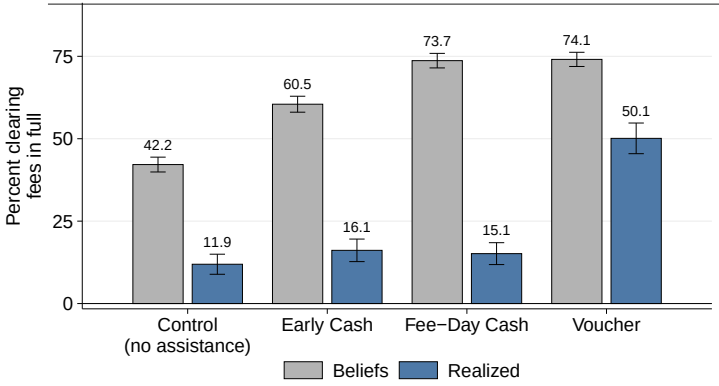
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by WTP model shocks

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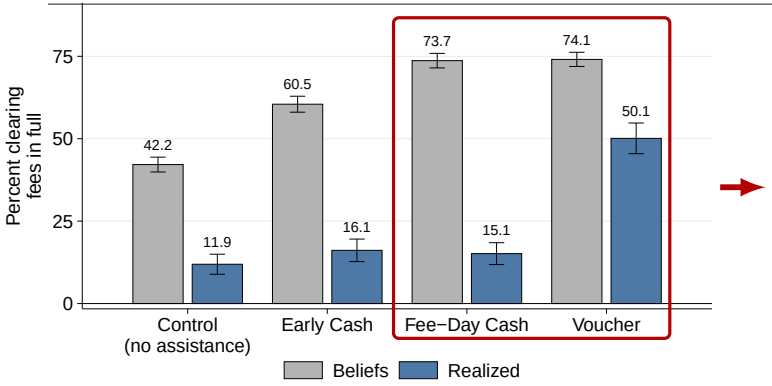


⇒ Parents are optimistic about clearing school fees under every transfer

⇒ Forecast errors can reflect mistakes about **circumstances** (Augenblick et al. 2026) or about **self** (Heidhues & Kőszegi 2010)

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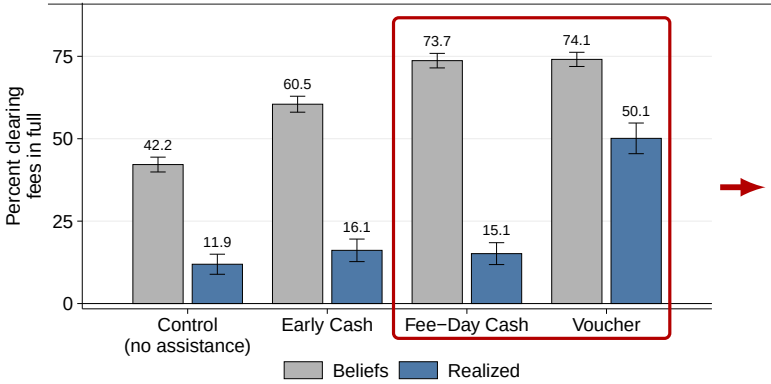


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How accurate are parents' forecasts of fee clearing?



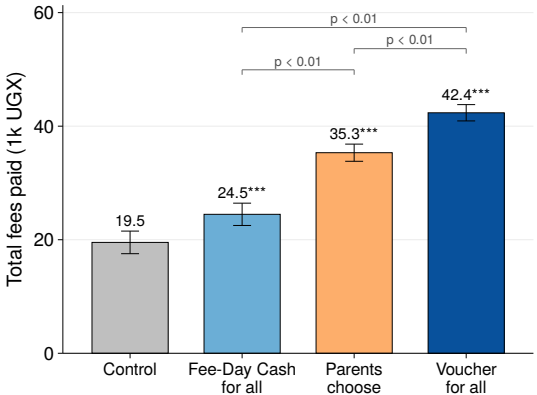
- Anticipated effect of earmarking: < 1 pp. Realized: $\sim$35 pp
  $\rightarrow$ parents underestimated the effect by $\sim$34 pp
- **Model interpretation:** consistent with partial naïveté ($\beta < \tilde{\beta} < 1$)

$\Rightarrow$ Parents are optimistic about clearing school fees under every transfer

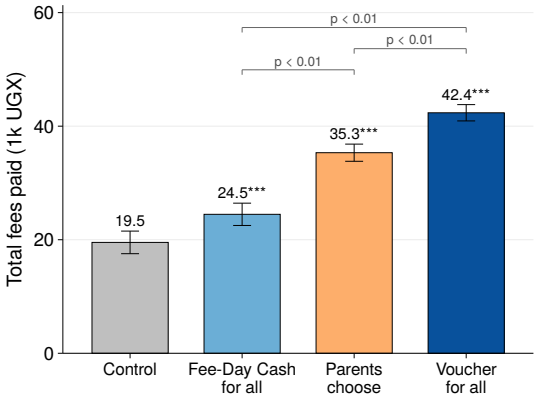
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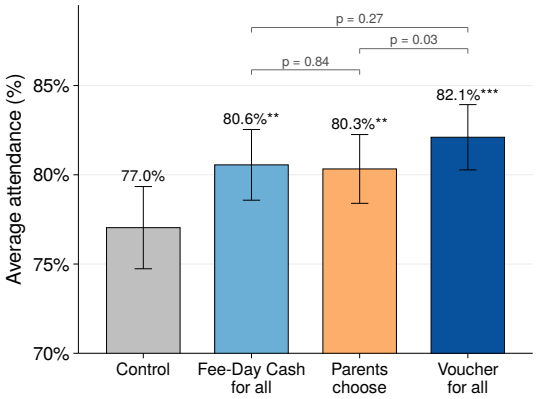
by WTP model shocks

Schooling outcomes under alternative assignment rules



Schooling outcomes under alternative assignment rules





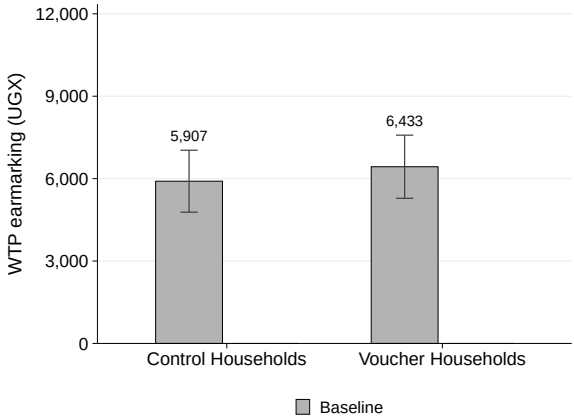
⇒ Universal voucher assignment outperforms demand-based choice on schooling outcomes

delay

Does experience change demand for earmarking?

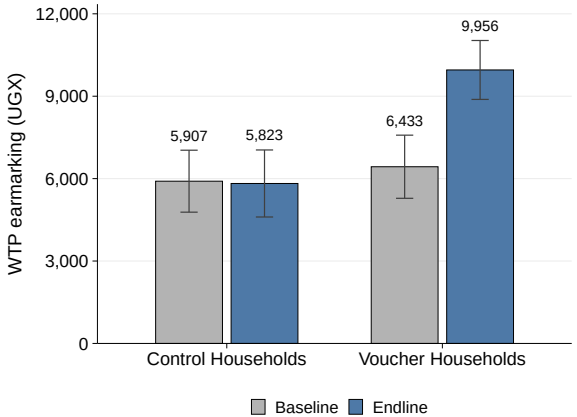
Does experience change demand for earmarking?

- Re-elicited Voucher vs. Fee-Day Cash at endline (incentivized), for Term 3 or Term 1
- If the Voucher locked parents in when they needed cash, WTP should **fall**



Does experience change demand for earmarking?

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⇒ Voucher recipients revise WTP up, Control flat → no sign of regret

table

Conclusion and policy takeaways

1. Parents have a high demand for commitment in education
2. All transfers raise fee payment; only the Voucher reduces dropout, and delaying cash adds little
3. Demand does not predict treatment effects on schooling outcomes

If the policy objective is schooling outcomes:

$\Rightarrow$ Pay the school, not the parent: same cost as cash, largest dropout reduction, and parents want it

$\Rightarrow$ Do not screen on parental demand: low-demand households can still have large schooling responses

Thank you!

Michelle Layvant

mlayvant@berkeley.edu

Baseline summary statistics

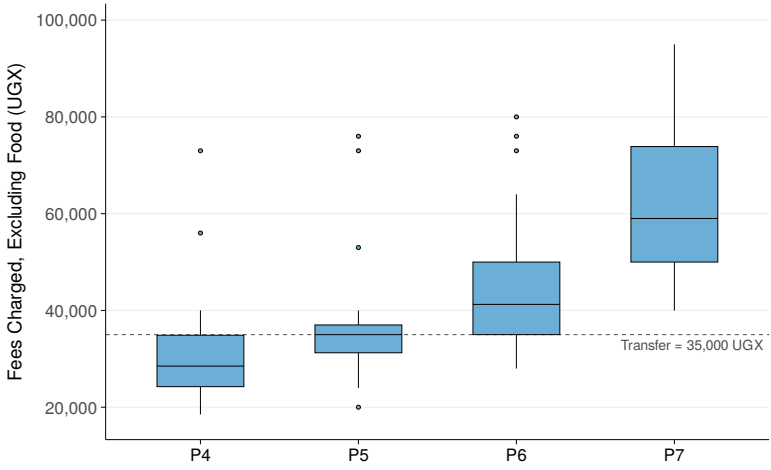
Child	Mean
Age (years)	13.2
In Primary 4 (rest Primary 6)	53%
Female	57%
Commute to school (minutes)	35
Term 1 fees charged, excl. food (UGX)	48,052

Primary guardian	Mean
Female	77%
Age (years)	44.0
Child's mother or father	67%
Completed primary (P7+)	45%
Household head	49%

Household	Mean
HH members	6.7
Adults	2.7
Children (<18)	4.0
Rooms in dwelling	2.8
Cement floor	47%
Single-adult household	16%
Income mainly agricultural	64%
Income, past 14 days (UGX)	84,478
Per capita	12,950
Electricity, any source (inc. solar)	55%
Any savings, past year	77%
Bank account	16%
SACCO or savings group	54%
Took a loan, past 12 months	57%

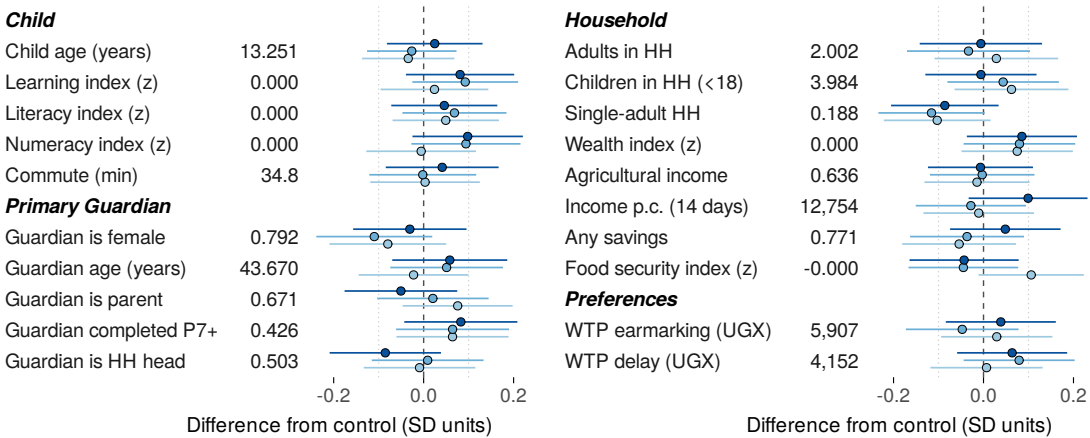
back balance

Term-by-term school fees increase by grade



back

Baseline balance across treatment arms

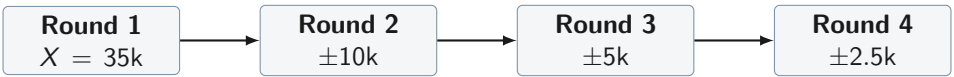


back

WTP elicitation: adaptive binary search

Task	Fixed option	Variable option
Earmarking	35k Voucher	X Fee-Day Cash
Delay	35k Early Cash	X Fee-Day Cash

Same adaptive procedure in both tasks:



Choose fixed 35k option $\Rightarrow$ raise X **Choose Fee-Day Cash** $\Rightarrow$ lower X

$\text{WTP}_{\text{earmark}} = X_V^* - 35{,}000$ $\text{WTP}_{\text{delay}} = 35{,}000 - X_D^*$

back

Incentive compatibility

- **2% of parents** receive a transfer according to their choices; treatment revealed *after the elicitation*
- **Becker–DeGroot–Marschak payout** (Allcott et al. 2022): draw one task and one price $X \in \{17.5\text{k}, 20\text{k}, \ldots, 52.5\text{k}\}$
  - X below her switch point $\rightarrow$ the fixed 35k option
  - X above her switch point $\rightarrow$ X in Fee-Day Cash
- Price drawn independently of her answers $\rightarrow$ misreporting never yields a better option
- Choice group excluded from the analysis sample (1,960 analyzed)

back

Each future self can break the plan

At $t = 0$ she maximizes $\beta\,\mathbb{E}[c_1+c_2+c_3] \Rightarrow$ everything is future, so she wants the fee paid

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Let $\tilde{\mu}_d$ denote the perceived probability of paying the fee under transfer d.

Fee-day self: "Will I pay?"

Choice: pay the fee or not

$$\max \quad c_2 + \tilde{\beta} c_3$$

$$\tilde{\beta} R \geq F \Rightarrow \textbf{pays}$$
$$\tilde{\beta} R < F \Rightarrow \textbf{does not pay}$$

Earmarking prevents $t = 2$ self from not paying

$$\text{WTP}_{\text{earmark}} = (R - F)(\tilde{\mu}_V - \tilde{\mu}_{FC}) = R - F$$

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Earlier self: "Will I save?"

Choice: savings, $s \geq 0$

$$\max \quad c_1 + \tilde{\beta}\,\mathbb{E}[c_2 + c_3]$$

$$\tilde{\beta} \geq \bar{\beta}_S \Rightarrow \textbf{saves}$$
$$\tilde{\beta} < \bar{\beta}_S \Rightarrow \textbf{spends Early Cash}$$

Delayed cash prevents $t = 1$ self from not saving

$$\text{WTP}_{\text{delay}} = (R - F)(\tilde{\mu}_{FC} - \tilde{\mu}_{EC}) = q(R - F)$$

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back saving WTP derivation

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$$\textbf{WTP} = (R - F)\Delta\tilde{\mu}$$

net return $\times$ perceived increase in the probability of paying

back saving WTP derivation

Saving decision under Early Cash

At $t = 1$ she chooses $s \in \{0,\ F - y_L\}$: save enough to pay in the low state, or nothing.

$$\max_{s \geq 0} \ \ c_1 + \tilde{\beta}\,\mathbb{E}[c_2 + c_3]$$

Gain from saving (given $\tilde{\beta}R \geq F$, so she pays whenever she can):

$$U_1(F - y_L) - U_1(0) = \underbrace{\tilde{\beta}\, q\,(R - F)}_{\text{avoids low-state failure}} - \underbrace{(1 - \tilde{\beta})(F - y_L)}_{\text{cost of carrying cash forward}}$$

Saves iff

$$\boxed{\tilde{\beta} \geq \bar{\beta}_S \equiv \frac{F - y_L}{(F - y_L) + q(R - F)}}$$

If $\tilde{\beta}R < F$, she would not pay even with cash in hand, so $s^* = 0$.

back

Value of each transfer at $t = 0$

At $t = 0$, all consumption is future:

$$V_0^{(d)} = \beta\,\mathbb{E}[c_1 + c_2 + c_3]$$

The resulting value functions are

$$V_0^{(EC)} = \beta\Big[y_1 + F + \mathbb{E}[y_2] + \tilde{\mu}_{EC}(R - F)\Big]$$

$$V_0^{(FC)} = \beta\Big[y_1 + F + \mathbb{E}[y_2] + \tilde{\mu}_{FC}(R - F)\Big]$$

$$V_0^{(V)} = \beta\Big[y_1 + F + \mathbb{E}[y_2] + \tilde{\mu}_V(R - F) - F(1 - \tilde{\mu}_V)\Big]$$

- $R - F$ is the net value when the fee clears.
- Unspent cash is kept; an unused Voucher is forfeited (but $\tilde{\mu}_V = 1$ in the benchmark).

WTP definition: x is the fee-day cash amount that makes her indifferent:

$$V_0^{(d)} = V_0^{(d')} + \beta x \quad \Rightarrow \quad x = \frac{V_0^{(d)} - V_0^{(d')}}{\beta}$$

next: derive WTP

back

Deriving willingness to pay

1. **WTP to delay cash** (Fee-Day Cash over Early Cash):

$$\begin{aligned}\text{WTP}_{\text{delay}} &\equiv \frac{V_0^{(FC)} - V_0^{(EC)}}{\beta} = (R-F)(\tilde{\mu}_{FC} - \tilde{\mu}_{EC}) \\ &= (R-F)\, q\, \mathbf{1}\{\tilde{\beta} R \geq F\} \mathbf{1}\{\tilde{\beta} < \bar{\beta}_S\} \\ &= \boxed{q(R-F)\, \mathbf{1}\{F/R \leq \tilde{\beta} < \bar{\beta}_S\}}\end{aligned}$$

2. **WTP to earmark** (Voucher over Fee-Day Cash):

$$\begin{aligned}\text{WTP}_{\text{earmark}} &\equiv \frac{V_0^{(V)} - V_0^{(FC)}}{\beta} = (R-F)(\tilde{\mu}_V - \tilde{\mu}_{FC}) - F(1-\tilde{\mu}_V) \\ &\underset{\tilde{\mu}_V = 1}{=} (R-F)\Big(1 - \mathbf{1}\{\tilde{\beta} R \geq F\}\Big) \\ &= \boxed{(R-F)\, \mathbf{1}\{\tilde{\beta} R < F\}}\end{aligned}$$

back

What if the Voucher only partially covers the fee?

In the experiment, $C < F$: the parent must top up the balance $F - C$ for the child to stay enrolled.

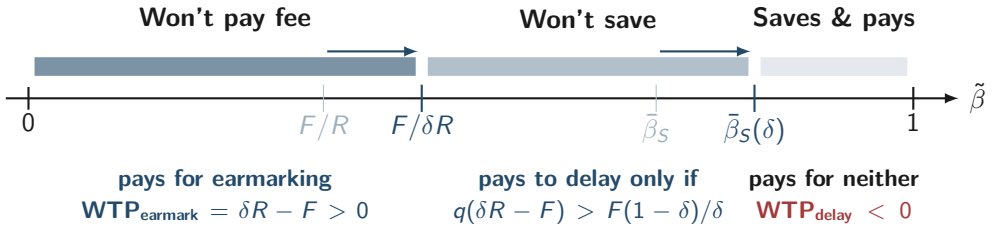
- **Paying gets easier.** She gives up $F - C$ instead of F to obtain $\tilde{\beta}R$. Parents with $\frac{F-C}{R} \leq \tilde{\beta} < \frac{F}{R}$ now pay.
- **Saving gets easier.** She needs to carry forward less, and failing to top up forfeits C: the return to saving rises, so $\bar{\beta}_S$ falls.
- **Not everyone gains.** If $\tilde{\beta}R < F - C$, she will not top up; C is lost and cash would have been better.

$\Rightarrow$ Partial earmarking lowers both thresholds: the "won't pay" region shrinks but does not disappear

back

Robustness: what if $\delta < 1$?

Pays iff $\tilde{\beta}\delta R \geq F$; saves iff $\tilde{\beta} \geq \bar{\beta}_S(\delta)$, with $\bar{\beta}_S(\delta) > \bar{\beta}_S$. Maintained: $\delta R > F$.



→ Earmarking: same date for both instruments, so δ cancels; $\text{WTP}_{\text{earmark}} > 0$ still requires $\tilde{\beta} < 1$.

→ Delay: waiting a period now costs $F(1-\delta)/\delta$, so time-consistent parents pay to get cash *earlier*.

⇒ Discounting cannot explain demand for earmarking and works against demand for delay

back

Robustness: what if utility is concave?

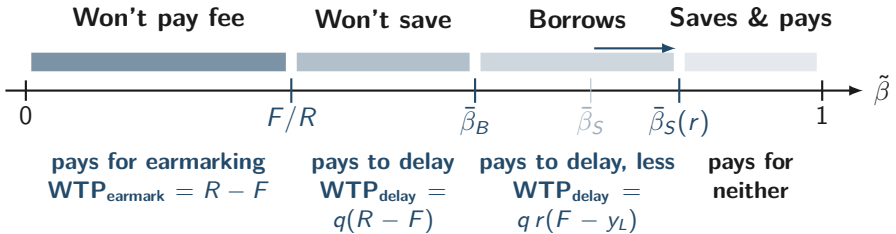
- **Paying depends on liquidity.** She pays iff $\tilde{\beta}\, u(R) \geq u(\ell) - u(\ell - F)$: richer households pay more readily, and saving is no longer all-or-nothing.
- **WTP is a money metric,** not simply $(R - F) \times \Delta\tilde{\mu}$: magnitudes and cutoffs change.
- **Revealed preference survives.** A time-consistent parent can replicate any restriction with flexible cash, so with costless saving $V_0^{\text{Early}} \geq V_0^{\text{FDC}} \geq V_0^{V}$: positive WTP still requires $\tilde{\beta} < 1$.

$\Rightarrow$ Concavity changes magnitudes and cutoffs, not the revealed-preference interpretation of positive WTP

back

Robustness: what if parents can borrow?

Fee loan rate r, repaid from R. Maintained: $(1+r)F \geq R$ (borrowing costs more than R)



→ Earmarking: an expensive loan never makes the $t=2$ self pay, so F/R is unchanged.

→ Delay: loans crowd out saving, and a borrower pays only to avoid interest, $q\,r(F-y_L) < q(R-F)$.

⇒ Costly borrowing leaves demand for earmarking unchanged and only attenuates demand for delay

back

Comparative statics: who demands commitment, and how much?

Extensive margin: who demands?

Voucher:

$$\tilde{\beta} < \frac{F}{R}$$

She expects her fee-day self **not to pay even with cash in hand**. A higher F/R makes payment harder to justify $\Rightarrow$ more demand for earmarking.

Delay:

$$\frac{F}{R} \leq \tilde{\beta} < \bar{\beta}_S$$

She expects to **pay if the cash survives**, but not to save enough. A lower y_L means more must be saved $\Rightarrow$ more demand for delay.

Intensive margin: how much? Conditional on remaining in the same region,

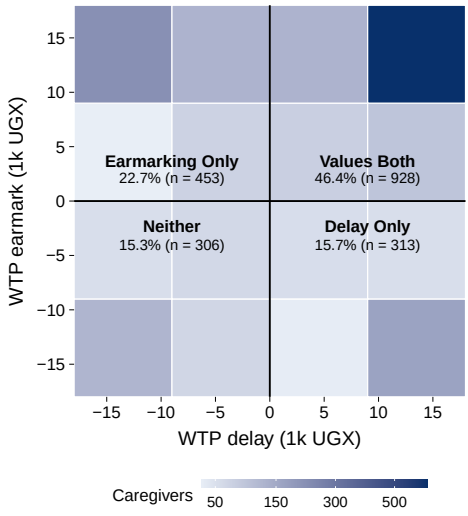
$$\boxed{\text{WTP}_{\text{earmark}} = R - F}, \qquad \boxed{\text{WTP}_{\text{delay}} = q(R - F)}.$$

The premium reflects the **cost of failure** and how often the restriction prevents it:

- Larger $R - F \Rightarrow$ larger premia.
- Higher $q \Rightarrow$ larger delay premium.
- q and y_L do not affect the earmarking premium.

wealth evidence | return evidence | back

Joint distribution of WTP



back

WTP estimation: two-limit Tobit

The elicited cash indifference amount is censored at UGX 17,500 and UGX 52,500 (equivalently, WTP is censored at $\pm$UGX 17,500):

$$\text{WTP}_i^* = Z_i'\pi + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$

$$\text{WTP}_i = \begin{cases} -17{,}500 & \text{if } \text{WTP}_i^* \leq -17{,}500, \\ \text{WTP}_i^* & \text{if } -17{,}500 < \text{WTP}_i^* < 17{,}500, \\ 17{,}500 & \text{if } \text{WTP}_i^* \geq 17{,}500. \end{cases}$$

back

Where would the cash have gone?

Caregivers who chose the Voucher at par, asked what the cash would have been used for

---

Stated destination of the cash

- Immediate household needs (food, medical care, essentials): **54%**
- Would have saved it: **fewer than 8%**
- Alcohol: 0.6%
- Support to relatives or others outside the household: 0.2%

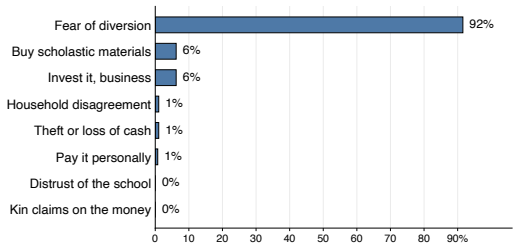
Open-ended explanations

- *"If it is sent on the phone, it might get other expenses, and I fail to pay fees."*
- *"I have a lot of problems, so I might use it for other business, and at the end fail to clear school fees when the term resumes."*

back

Fear of diverting the cash is the dominant stated reason for choosing the voucher

Reasons given in open-ended explanations, immediately after the incentivized choice

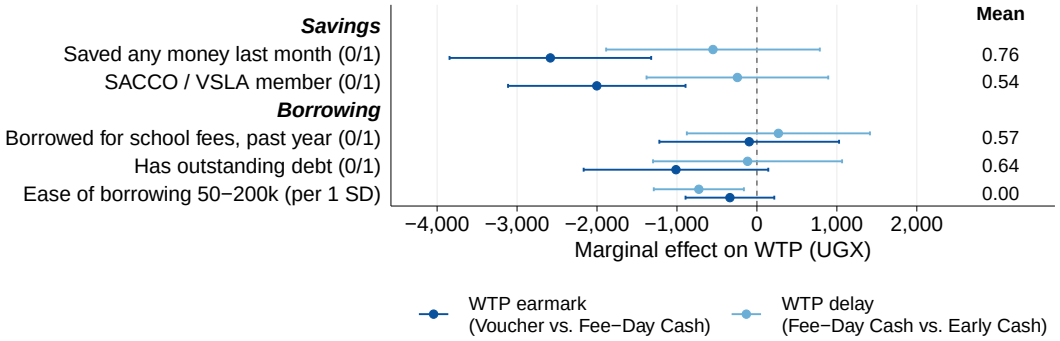


Regex dictionary coding. Share of the 675 of 1,381 Voucher-choosers whose words matched at least one reason; the other 51% matched none. Categories overlap.

- *"If it is sent on the phone, it might get other expenses, and I fail to pay fees."*
- *"I prefer subsidy because it's safe for me not to divert it in to other usages"*
- *"To avoid temptation of diverting that money to other household needs like food, medical, and many others"*

$\Rightarrow$ The alternative mechanisms are not raised often: intra-household disagreement, kin taxes, and fear of theft together account for 13 of 675 responses.

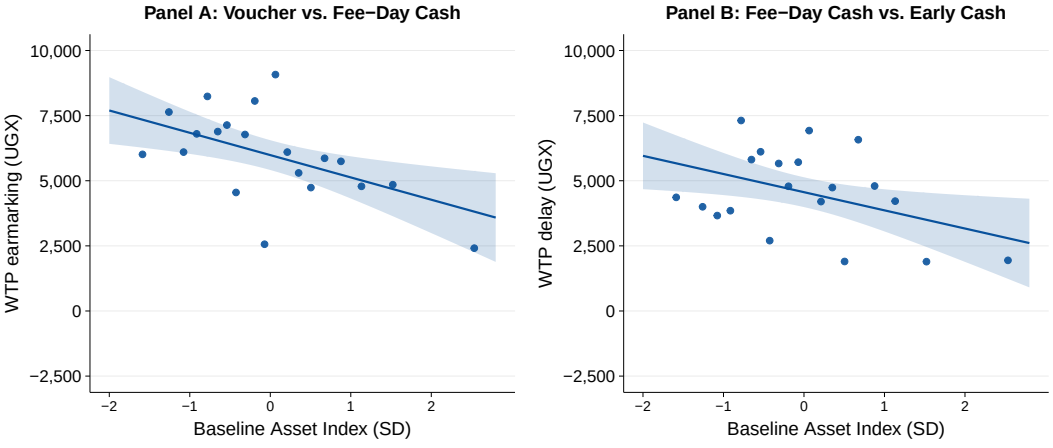
Demand for delay is not explained by a savings constraint



⇒ Parents without a place to save do **not** pay more to delay; savings vehicles lower the **earmark** premium instead

borrowing model back

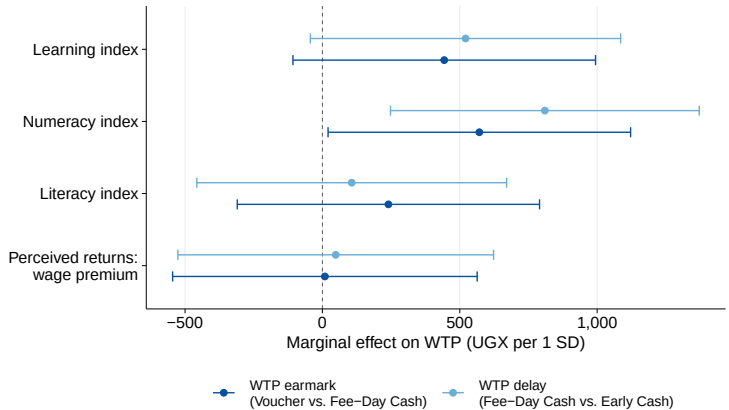
Both premia are decreasing in wealth



comparative statics back

Premia rise with numeracy, not with stated returns to education

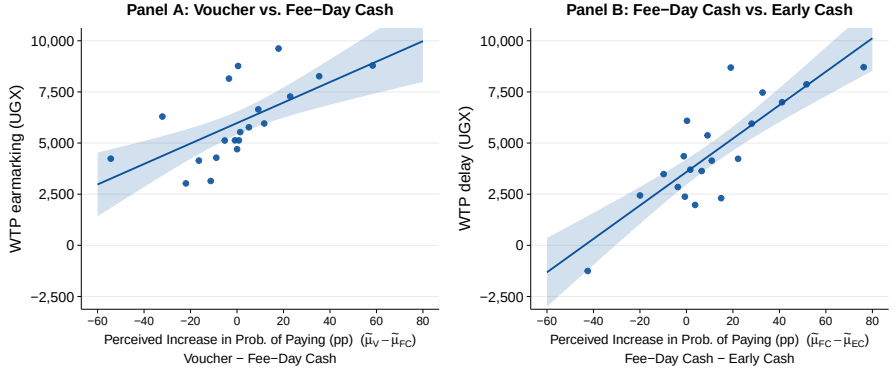
Model: conditional on demanding commitment, both premia scale with the net return $R - F$



Perceived returns: caregiver's expected earnings gap at age 25 between completing Senior 4 and Primary 7, conditional on working (Attanasio & Kaufmann).

comparative statics back

WTP rises with perceived increase in payment probability: controlling for net returns (R-F)



Both axes residualized on baseline controls: grade, learning, numeracy and literacy indices, and perceived returns to education. Slopes per 10 pp: 501 UGX (Panel A), 817 UGX (Panel B), vs. 481 and 821 without controls.

back

Does an eight-day shift in the payment date change behavior?

Within Fee-Day Cash, 90 (of 490) parents randomly offered a binding Feb 2 vs. Feb 10 choice:

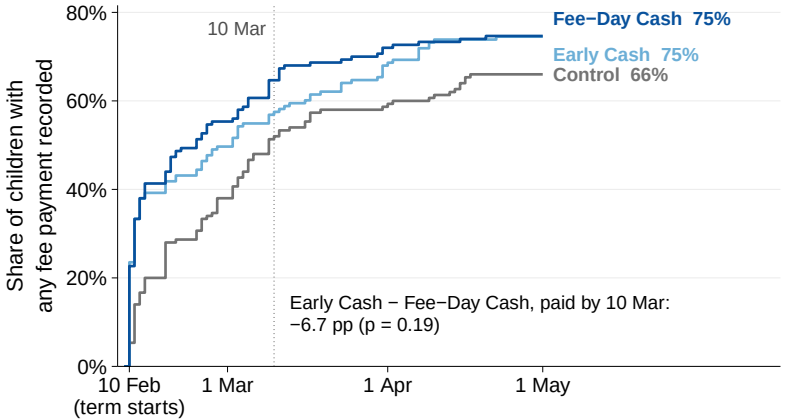
$$Y_i = \alpha + \tau \underbrace{\mathbf{1}\{\text{Paid Feb 10}\}_i}_{\text{instrumented by } \mathbf{1}\{\text{Offered choice}\}_i} + \gamma_{s(i)} + \varepsilon_i$$

	Total Fees Paid (1k UGX)	Share Fees Paid	Attendance
Paid on February 10 (IV)	-3.607	-0.043	0.021
	(3.786)	(0.076)	(0.035)
February-2 Mean	24.367	0.477	0.805
Observations	446	446	5,101
First stage: share paid Feb 10 | offered	0.671	0.671	0.671

$\Rightarrow$ Aligning cash with the fee day does not detectably change payment or attendance (LATE for those who choose delay; imprecise)

back

Early Cash delays payment slightly relative to Fee-day Cash (n.s.)



Notes: Share of children with any Term 1 fee payment (cash or in-kind) recorded on or before each date; 11 schools that returned daily fee registers, non-transfers, $N = 611$.

back

Estimating equation: intention-to-treat effects

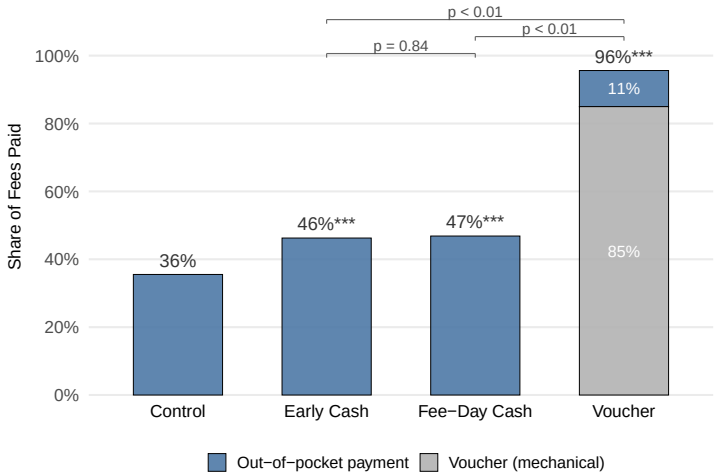
$$Y_{is} = \beta_1 \, T_{1,is} + \beta_2 \, T_{2,is} + \beta_3 \, T_{3,is} + X'_{is}\gamma + \delta_s + \varepsilon_{is}$$

- T_1 Early Cash, T_2 Fee-Day Cash, T_3 Voucher; Control omitted
- δ_s: strata FE (school $\times$ grade $\times$ gender)
- X_{is}: baseline outcome where available (ANCOVA)
- Robust SEs; weekly attendance: week FE, clustered by child
- 1,960 parents (Choice group excluded); Anderson (2008) sharpened q-values across 5 primary outcomes

$\Rightarrow$ $\beta_3 - \beta_2$: **earmarking**; $\beta_2 - \beta_1$: **timing**; $\beta_3 - \beta_1$: **both restrictions**

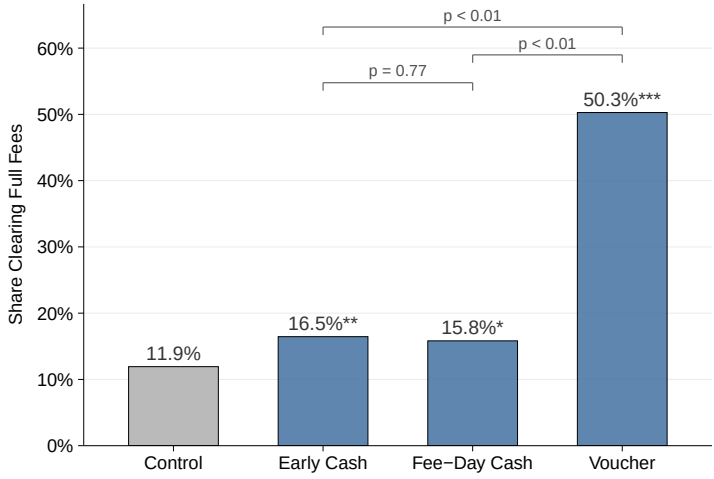
back

Share of fees paid



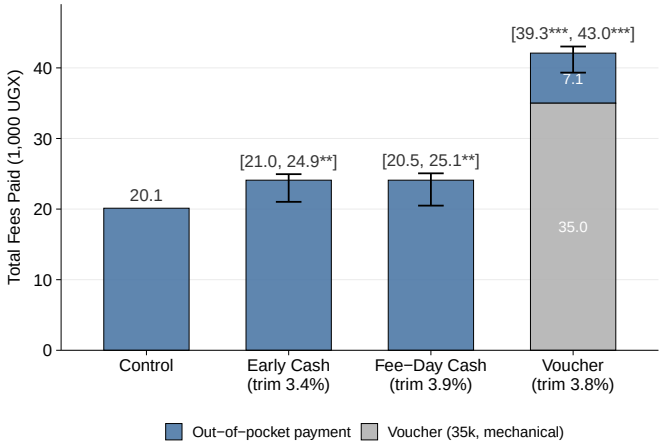
back

Share paid fees in full



back

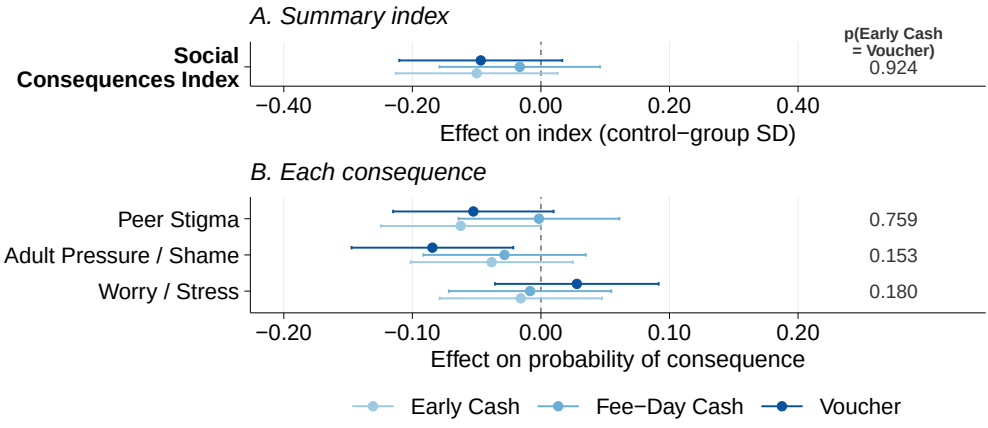
Fee payment (Lee bounds)



Notes: Bars are raw mean fees paid (1k UGX) among children who did not transfer schools, zero where no payment was recorded; the Voucher bar includes the 35k paid to the school. Non-transfer rate is 88% in control vs. 91–92% in each arm. Whiskers are Lee (2009) bounds on the always-stayer mean, trimming each treated arm by its excess non-transfer share from the top or bottom . Lower-bound effects vs. control: Voucher +19.2k (bootstrap s.e. 2.0), Fee-Day Cash +0.4k (2.4), Early Cash +0.9k (2.5). Choice group excluded.

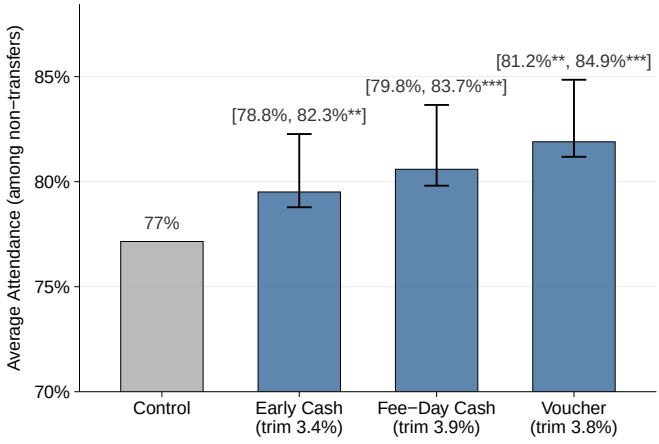
back

Social consequences for late or unpaid fees (as reported by the index child)



back

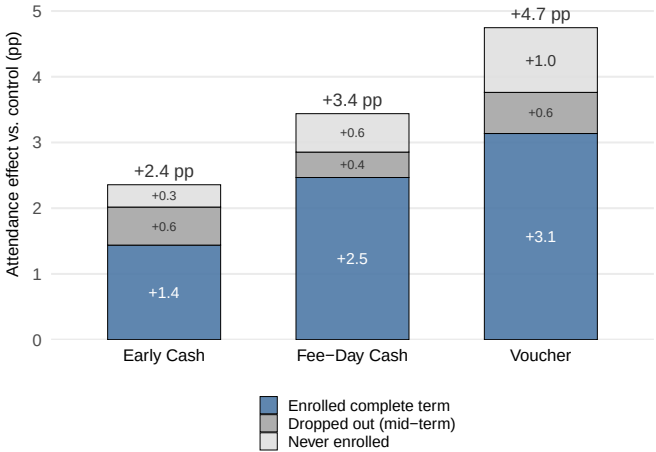
Attendance (Lee bounds)



Notes: Bars are raw mean attendance among children who did not transfer schools; dropouts count as zero. Non-transfer rate is 88% in control vs. 91–92% in each arm, so treated samples are positively selected. Whiskers are Lee (2009) bounds on the always-stayer mean: each treated arm's distribution is trimmed by its excess non-transfer share from the top or bottom. Lower-bound effects vs. control: Voucher +4.0 pp (bootstrap s.e. 1.6), Fee-Day Cash +2.7 (1.6), Early Cash +1.6 (1.7). Choice group excluded.

back

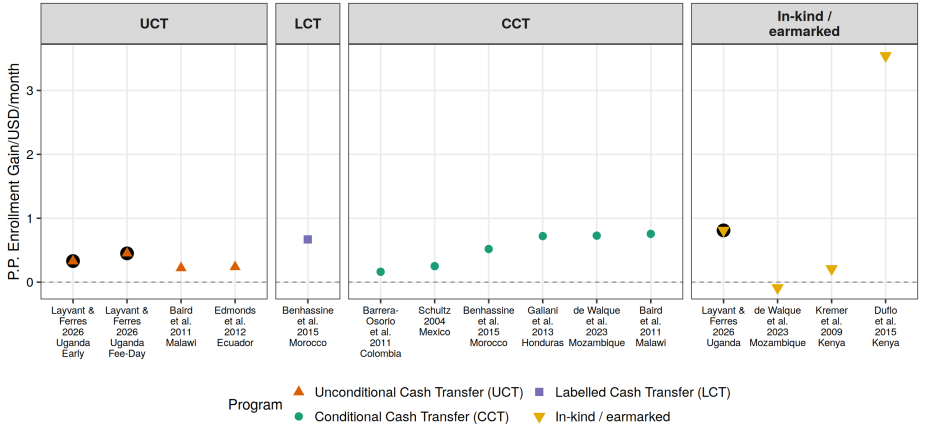
Attendance decomposition



- Attendance in each arm is a share-weighted average over three enrollment states: enrolled all term, dropped out mid-term, never enrolled
- Split the treatment–control gap into (i) children moving to a higher-attendance state and (ii) higher attendance within a state; midpoint weights make the pieces sum exactly to the total (Kitagawa, 1955)

back

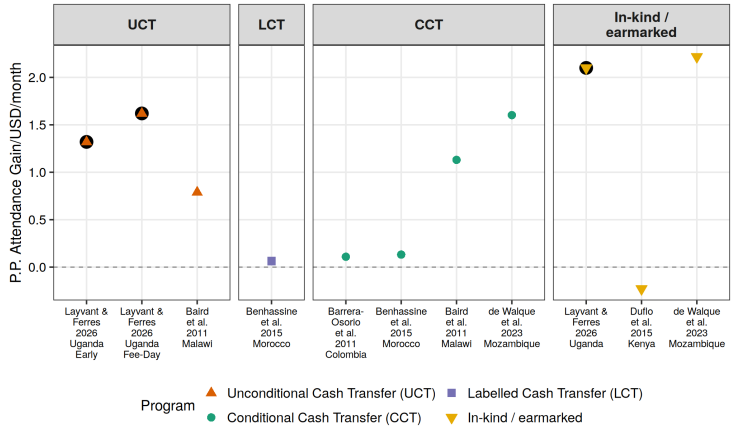
Enrollment comparison with other RCTs



ITT effect divided by the average monthly transfer per child (García & Saavedra 2017). Administrative costs excluded.

back

Attendance comparison with other RCTs



ITT effect divided by the average monthly transfer per child (García & Saavedra 2017). Administrative costs excluded.

back

Spillovers within the school

Randomization is within school × grade × sex strata, so treated and control children share classrooms. Tests use daily attendance from the class registers, which cover every child in the class.

- **Onto control children** (e.g., discouragement)
  Control study children attend like classmates who were never in the study, and the gap does not build over the term
- **Classroom-level** (peer effects, teacher response)
  Classmates' attendance does not vary with the share of their class that was treated, or with the share holding a voucher
- **School-level response** (e.g., relaxed fee enforcement for everyone)
  Not testable: every school had the same number of treated children, and there is no pre-treatment term to compare against. If schools eased up on everyone, unpaid children gain most, so every comparison is biased toward zero

back

No gains in Uwezo literacy and numeracy assessments

	(1) Learning Index	(2) English Index	(3) Math Index
T1: Early Cash	0.059	0.003	0.096*
	(0.044)	(0.043)	(0.052)
	[0.158]		
T2: Fee-Day Cash	0.015	0.009	0.030
	(0.045)	(0.044)	(0.051)
	[0.329]		
T3: Voucher	0.000	-0.006	0.020
	(0.044)	(0.044)	(0.051)
	[0.362]		
Control Mean	0.000	0.000	0.000
Observations	1,956	1,956	1,959
p-value: T1 = T2	0.327	0.903	0.209
p-value: T2 = T3	0.743	0.734	0.846
p-value: T1 = T3	0.180	0.818	0.140

Notes: Uwezo assessments administered to the child at endline. Each index is z-scored within baseline grade on the endline control group; the learning index averages the English and math indices. ANCOVA on the matching baseline index; strata FE, robust SEs in parentheses. Brackets: Anderson (2008) sharpened q-values across the 15 primary tests. Choice group excluded.

back

No gains in school test scores

	Total Marks	English	Math	Science	Social Studies
Panel A: Mid-Term Assessment (T1 2026)					
T1: Early Cash	-0.087	-0.012	-0.011	-0.072	-0.057
	(0.078)	(0.080)	(0.075)	(0.079)	(0.082)
T2: Fee-Day Cash	-0.052	0.009	0.001	0.053	-0.028
	(0.079)	(0.082)	(0.076)	(0.079)	(0.083)
T3: Voucher	-0.078	-0.036	-0.002	0.021	-0.100
	(0.077)	(0.083)	(0.078)	(0.078)	(0.080)
Control Mean	0.000	0.000	0.000	0.000	0.000
Observations	869	860	860	857	843
Panel B: End-of-Term Assessment (T1 2026)					
T1: Early Cash	-0.058	-0.037	-0.039	-0.053	-0.033
	(0.062)	(0.068)	(0.059)	(0.059)	(0.065)
T2: Fee-Day Cash	-0.032	0.035	-0.040	-0.019	-0.013
	(0.060)	(0.075)	(0.057)	(0.057)	(0.060)
T3: Voucher	-0.020	-0.006	-0.003	-0.020	-0.011
	(0.060)	(0.064)	(0.059)	(0.057)	(0.062)
Control Mean	0.000	0.000	0.000	0.000	0.000
Observations	1,570	1,569	1,565	1,566	1,557

Notes: School-administered mid-term (Panel A) and end-of-term (Panel B) exams, Term 1 2026, from school records. Scores are z-scored within 2026 grade on the control group. Samples are children who sat the exam; the mid-term was collected from a subset of schools. ITT with strata FE; robust SEs in parentheses; no baseline exam, so no ANCOVA. Choice group excluded.

back

Minimal gains in child socio-emotional wellbeing

	Depression (CES-DC)	Resilience Scale
T1: Early Cash	0.031	0.125**
	(0.062)	(0.060)
T2: Fee-Day Cash	0.032	0.016
	(0.062)	(0.062)
T3: Voucher	-0.052	0.023
	(0.065)	(0.065)
Control Mean	0.000	0.000
Observations	1,943	1,953
p-value: T1 = T2	0.999	0.066
p-value: T2 = T3	0.200	0.915
p-value: T1 = T3	0.201	0.101

CES-DC (child self-report)

- 20 depressive-symptom items, past week, each 0–3; score 0–60
- Reversed and z-scored on the endline control group: higher = fewer symptoms

Resilience scale (child self-report)

- 11 CYRM-12 items (e.g. "I feel that I belong at my school", "My family cares about me when times are hard"), each 1–5
- Summed and z-scored on the endline control group

Both ANCOVA on the baseline score, strata FE

back

Heterogeneity by child gender

	Total fees paid (1k UGX)			Dropout (pp)			Attendance (pp)			Learning index (z)		
	Boys	Girls	$p(=)$	Boys	Girls	$p(=)$	Boys	Girls	$p(=)$	Boys	Girls	$p(=)$
T1: Early Cash	5.4***	4.0**	0.63	-3.0	0.2	0.24	0.2	4.4**	0.19	0.07	0.05	0.89
	(2.1)	(2.0)		(2.3)	(1.5)		(2.4)	(2.1)		(0.07)	(0.06)	
T2: Fee-Day Cash	4.8**	4.8**	1.00	-3.8*	0.1	0.15	2.7	3.8*	0.71	0.06	-0.02	0.40
	(2.1)	(2.0)		(2.3)	(1.5)		(2.4)	(2.1)		(0.07)	(0.06)	
T3: Voucher	22.7***	22.9***	0.93	-4.4**	-1.5	0.25	2.0	7.4***	0.08	-0.04	0.03	0.42
	(1.9)	(1.7)		(2.2)	(1.3)		(2.3)	(2.0)		(0.06)	(0.06)	
Control mean	19.6	20.6		7.7	3.1		76.0	78.1		-0.12	0.08	
Observations	791	985		849	1,111		791	985		846	1,110	

Each arm interacted with the subgroup indicator; columns report the arm effect within each group, $p(=)$ tests equality across groups. Strata FE, robust SE.

back

Heterogeneity by child grade

	Total fees paid (1k UGX)			Dropout (pp)			Attendance (pp)			Learning index (z)		
	P6	P4	$p(=)$	P6	P4	$p(=)$	P6	P4	$p(=)$	P6	P4	$p(=)$
T1: Early Cash	4.4*	4.9***	0.87	-1.8	-0.6	0.65	3.7	1.4	0.47	-0.00	0.11*	0.19
	(2.6)	(1.4)		(2.1)	(1.7)		(2.6)	(2.0)		(0.06)	(0.07)	
T2: Fee-Day Cash	4.5*	5.0***	0.87	-0.2	-2.8*	0.34	2.4	4.2**	0.58	-0.01	0.04	0.55
	(2.6)	(1.4)		(2.2)	(1.5)		(2.6)	(1.9)		(0.06)	(0.07)	
T3: Voucher	18.5***	26.5***	0.00	-2.3	-3.1**	0.73	5.9**	4.3**	0.60	-0.02	0.01	0.73
	(2.4)	(1.1)		(2.0)	(1.4)		(2.3)	(1.9)		(0.06)	(0.06)	
Control mean	30.7	10.5		6.0	4.2		76.5	77.7		0.00	0.00	
Observations	839	937		925	1,035		839	937		924	1,032	

Each arm interacted with the subgroup indicator; columns report the arm effect within each group, $p(=)$ tests equality across groups. Strata FE, robust SE.

back

Heterogeneity by whether primary guardian is child's mother

	Total fees paid (1k UGX)			Dropout (pp)			Attendance (pp)			Learning index (z)		
	Other CG	Mother	$p(=)$	Other CG	Mother	$p(=)$	Other CG	Mother	$p(=)$	Other CG	Mother	$p(=)$
T1: Early Cash	6.4***	2.9	0.22	-3.3*	0.7	0.12	-0.6	5.5**	0.06	0.13**	-0.01	0.11
	(2.0)	(2.1)		(1.9)	(1.8)		(2.2)	(2.3)		(0.06)	(0.06)	
T2: Fee-Day Cash	5.8***	3.7*	0.48	-1.7	-1.5	0.95	1.1	5.6**	0.16	0.03	0.00	0.80
	(2.0)	(2.1)		(1.9)	(1.7)		(2.1)	(2.3)		(0.06)	(0.06)	
T3: Voucher	24.3***	21.3***	0.25	-3.4*	-2.1	0.59	2.5	7.5***	0.11	0.05	-0.05	0.28
	(1.8)	(1.9)		(1.9)	(1.5)		(2.2)	(2.2)		(0.06)	(0.06)	
Control mean	18.6	21.6		5.9	4.3		78.7	75.7		-0.06	0.06	
Observations	883	893		971	989		883	893		968	988	

Each arm interacted with the subgroup indicator; columns report the arm effect within each group, $p(=)$ tests equality across groups. Strata FE, robust SE.

back

Heterogeneity by household wealth

	Total fees paid (1k UGX)			Dropout (pp)			Attendance (pp)			Learning index (z)		
	Below med.	Above med.	$p(=)$	Below med.	Above med.	$p(=)$	Below med.	Above med.	$p(=)$	Below med.	Above med.	$p(=)$
T1: Early Cash	5.3**	3.9**	0.65	0.2	-2.7	0.29	2.6	2.5	0.97	0.06	0.06	1.00
	(2.1)	(2.0)		(2.0)	(1.8)		(2.3)	(2.2)		(0.06)	(0.06)	
T2: Fee-Day Cash	5.2**	4.1**	0.72	-0.9	-2.3	0.58	2.3	4.3*	0.51	-0.05	0.07	0.21
	(2.1)	(2.0)		(1.8)	(1.9)		(2.2)	(2.2)		(0.07)	(0.06)	
T3: Voucher	24.1***	21.5***	0.32	-2.5	-3.0	0.86	4.3**	5.6***	0.66	-0.02	0.01	0.74
	(1.8)	(1.8)		(1.6)	(1.8)		(2.2)	(2.1)		(0.06)	(0.06)	
Control mean	20.4	19.8		4.9	5.2		76.4	77.9		0.01	-0.01	
Observations	879	893		977	979		879	893		975	977	

Each arm interacted with the subgroup indicator; columns report the arm effect within each group, $p(=)$ tests equality across groups. Strata FE, robust SE.

back

HH budget: education spending, transfers out, food security

	(1) Total HH Schooling Spending (UGX)	(2) Net Transfers Out (UGX)	(3) Food Insecurity (FIES)
T1: Cash Today	35236**	10096**	-0.115*
	(16506)	(4924)	(0.062)
T2: Cash Fee-Day	15842	2600	-0.172***
	(15818)	(4933)	(0.065)
T3: Voucher	38095**	1188	-0.123*
	(16251)	(5488)	(0.066)
Control Mean	288000	-6804	0.000
Observations	1,960	1,939	1,960
T1 Effect (%)	12.2		
T2 Effect (%)	5.5		
T3 Effect (%)	13.2		
p-value: T1 = T2	0.230	0.130	0.380
p-value: T2 = T3	0.159	0.794	0.471
p-value: T1 = T3	0.863	0.101	0.911

Notes: Parent endline survey. OLS with strata FE; robust SEs in parentheses; cols. (2) and (3) are ANCOVA on the baseline value. Col. (1) is out-of-pocket spending on all children in the household plus the 35,000 voucher for Voucher children enrolled at the baseline school. Col. (2) is money given minus received from other households (winsorized). Col. (3), FIES: 8 yes/no items summed 0–8, z-scored on the control group (e.g. "You were worried you would not have enough food to eat", "You were hungry but did not eat"); higher = more insecure; control mean 6.1 of 8. Choice group excluded.

back

Household education spillovers (shares excluding the index child)

	(1) Share Attending School	(2) Share Sent Home over Fees	(3) Share w/ School Feeding
T1: Early Cash	-0.018	-0.017	0.049*
	(0.012)	(0.025)	(0.029)
T2: Fee-Day Cash	-0.014	-0.068***	0.051*
	(0.011)	(0.026)	(0.029)
T3: Voucher	-0.018	-0.020	-0.012
	(0.011)	(0.024)	(0.029)
Control Mean	0.961	0.771	0.545
Observations	1,802	1,766	1,767
T1 Effect (%)	-1.9	-2.2	9.0
T2 Effect (%)	-1.5	-8.8	9.4
T3 Effect (%)	-1.9	-2.6	-2.2
p-value: T1 = T2	0.754	0.056	0.948
p-value: T2 = T3	0.747	0.065	0.030
p-value: T1 = T3	1.000	0.907	0.033

Notes: Parent endline survey. Outcomes are shares over the household's school-age (6–17) siblings of the index child: col. (1) currently attending school in 2026; col. (2) sent home over unpaid fees since January 2026 and col. (3) receiving school feeding, both among siblings currently attending. Households with no school-age sibling are excluded. OLS with strata FE; robust SEs in parenthese. Choice group excluded.

back

Why the forecast error implies partial naïveté

Setup. $\mu_d(\beta)$ = prob. of clearing fees under $d \in \{V, FC\}$; she forecasts with perceived $\tilde{\beta}$: $\tilde{\mu}_d = \mu_d(\tilde{\beta}) + \varepsilon_d$, with $\varepsilon_V = \varepsilon_{FC}$ (shocks hit both arms alike). Earmark effect $G(\beta) \equiv \mu_V(\beta) - \mu_{FC}(\beta)$, decreasing in β.

1. Forecast error signs $\tilde{\beta} - \beta$

$$(\tilde{\mu}_V - \tilde{\mu}_{FC}) - (\mu_V - \mu_{FC}) = G(\tilde{\beta}) - G(\beta) < 0 \iff \tilde{\beta} > \beta$$

Data: $0.5 - 34.5 < 0 \Rightarrow \tilde{\beta} > \beta$

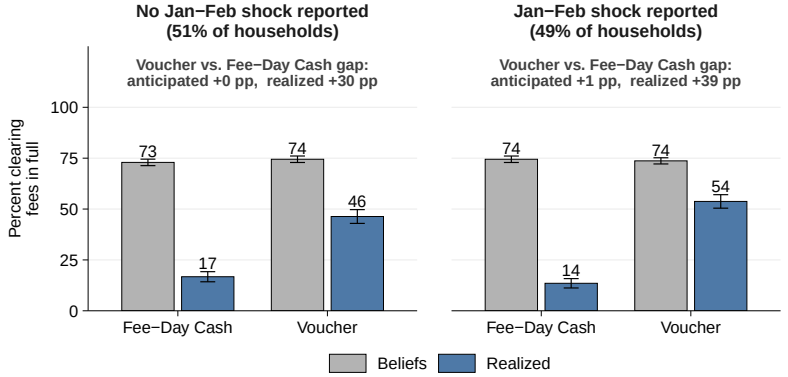
2. Demand for the earmark signs $\tilde{\beta} - 1$

$$\text{WTP}_{\text{earmark}} = (R - F)(1 - \tilde{\mu}_{FC}) > 0 \iff \tilde{\beta} < 1$$

$\Rightarrow \beta < \tilde{\beta} < 1$**: partial naïveté.** In probabilities: $\mu_{FC} < \tilde{\mu}_{FC} < 1$, data $0.15 < 0.73 < 1$.

back

The forecast error is not explained by shocks



Shock reported for January–February: 49% of households, balanced across arms.

⇒ Cash goes unpaid without a shock: the error is her own use of cash in hand, not the state of the world

back

Treatment effects on schooling decline in WTP

	(1) Total Paid (1k UGX)	(2) Share Paid	(3) Cleared Fees	(4) Attendance
T1: Early Cash	3.259**	0.099***	0.043*	0.021
	(1.606)	(0.032)	(0.023)	(0.019)
T2: Fee-Day Cash	3.726**	0.097***	0.031	0.023
	(1.556)	(0.030)	(0.022)	(0.018)
T3: Voucher	23.310***	0.626***	0.411***	0.053***
	(1.370)	(0.029)	(0.027)	(0.017)
T1 × WTP earmark (1k UGX)	0.206*	0.002	0.000	0.001
	(0.112)	(0.002)	(0.002)	(0.001)
T2 × WTP earmark (1k UGX)	0.186*	0.003	0.001	0.002
	(0.106)	(0.002)	(0.002)	(0.001)
T3 × WTP earmark (1k UGX)	-0.073	-0.005**	-0.004**	-0.000
	(0.096)	(0.002)	(0.002)	(0.001)
Slope: Voucher − Fee-Day Cash	-0.258	-0.007	-0.005	-0.002
	(0.088)	(0.002)	(0.002)	(0.001)
p-value	0.003	0.000	0.004	0.049
Control Mean	20.123	0.374	0.119	0.772
Observations	1,776	1,776	1,776	1,776

Notes: Each column regresses the outcome on the three arm indicators, WTP earmark, and the three arm × WTP interactions, with strata FE and robust SEs. WTP earmark is the baseline indifference value of the voucher minus 35,000, in 1k UGX (range −17.5 to 17.5). The arm rows are treatment effects for a parent with WTP = 0. Attendance is the spot-check average (share of visits present; dropouts at zero); dropout from the parent endline. Non-transfer sample; choice group excluded.

Slope = (T3 × WTP) − (T2 × WTP): how the effect of the earmark itself changes with WTP.

→ If parents sorted on their true treatment effects, it would be positive.

back

Beliefs do not predict treatment effects on schooling

	(1) Total Paid (1k UGX)	(2) Share Paid	(3) Cleared Fees	(4) Attendance
T1: Early Cash	4.772***	0.118***	0.043**	0.026
	(1.407)	(0.028)	(0.021)	(0.016)
T2: Fee-Day Cash	4.822***	0.112***	0.039*	0.034**
	(1.428)	(0.028)	(0.021)	(0.016)
T3: Voucher	22.851***	0.596***	0.384***	0.051***
	(1.260)	(0.026)	(0.024)	(0.015)
T1 × Perceived Δ Prob. Paying (10 pp)	-1.007*	-0.016	0.006	-0.009
	(0.600)	(0.012)	(0.009)	(0.006)
T2 × Perceived Δ Prob. Paying (10 pp)	-0.385	-0.006	0.001	-0.006
	(0.579)	(0.011)	(0.008)	(0.006)
T3 × Perceived Δ Prob. Paying (10 pp)	-0.710	-0.008	-0.006	-0.010*
	(0.505)	(0.010)	(0.010)	(0.006)
Slope: Voucher − Fee-Day Cash	-0.325	-0.002	-0.006	-0.004
	(0.524)	(0.011)	(0.010)	(0.006)
p-value	0.535	0.830	0.552	0.498
Control Mean	20.123	0.374	0.119	0.772
Observations	1,776	1,776	1,776	1,776

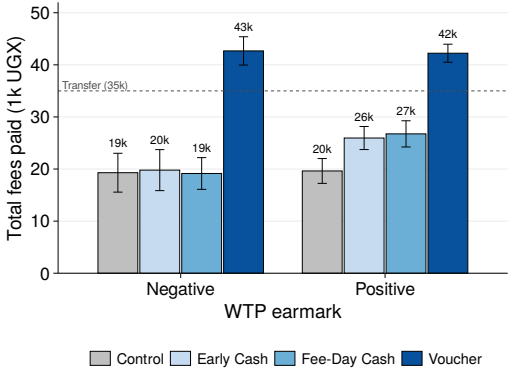
Notes: Same specification as the WTP table with the stated belief in place of WTP. Perceived Δ Prob. Paying $= \tilde{\mu}_V - \tilde{\mu}_{FC}$ is the parent's baseline forecast of the probability of fully paying under the voucher minus the same under fee-day cash, in 10 pp (range −10 to 10). The arm rows are treatment effects for a parent who expects no difference. Attendance is the spot-check average (share of visits present; dropouts at zero); dropout from the parent endline. Non-transfer sample; choice group excluded. Strata FE, robust SEs in parentheses.

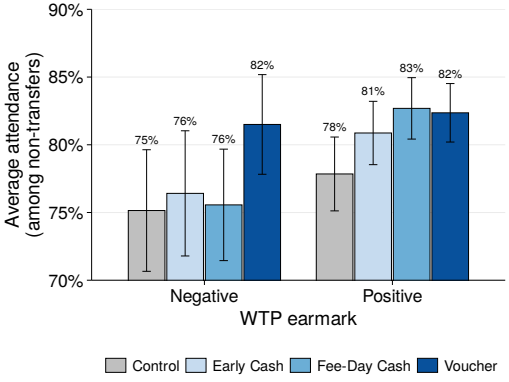
Slope = (T3 × belief) − (T2 × belief): how the effect of the earmark itself changes with the perceived difference.

→ Under the model with correct forecasts ($\tilde{\mu}_d = \mu_d$), the slope in col. (3) would be +0.10: a 10-pp higher forecast, a 10-pp larger effect.

back

Fees paid and attendance by WTP for the earmark, all arms



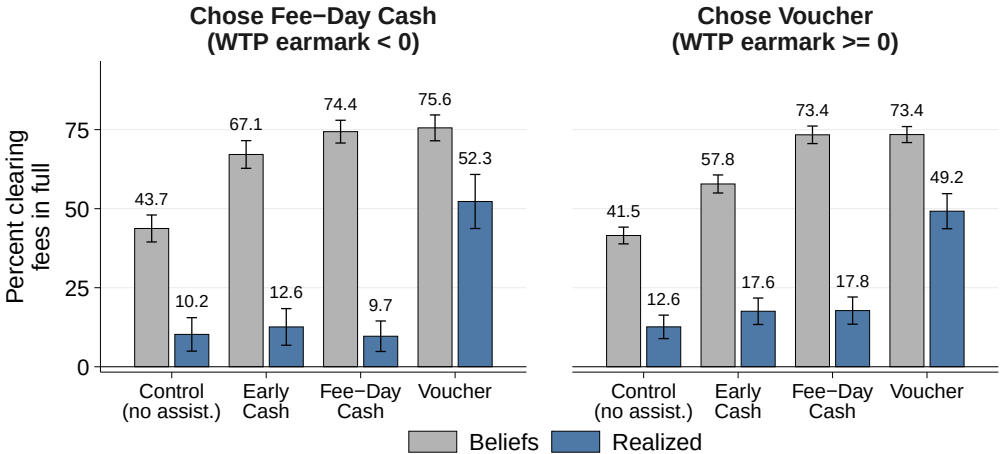


⇒ **Chose Fee-day Cash:** cash leaves fees and attendance at control; the Voucher adds 23k in fees and 6 pp in child's attendance

⇒ **Chose the Voucher:** cash raises fees 7k and attendance as much as the Voucher

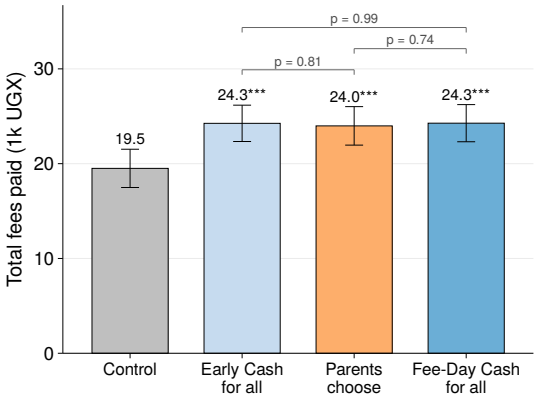
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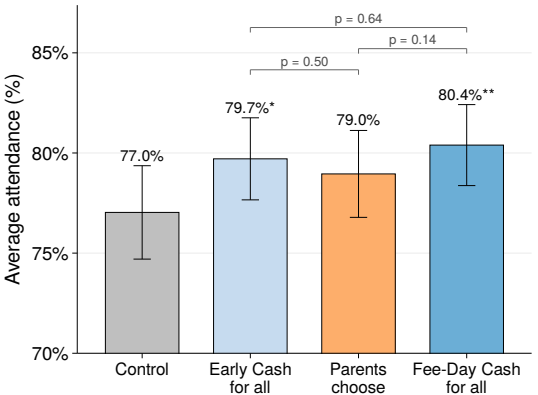
Forecasts and clearing by baseline choice, all arms



back

Assignment rules on the timing margin





back

Effects of experience on voucher demand

	(1) Full Sample	(2) Same Caregiver	(3) Term 1 Only	(4) Same CG, Term 1
T1: Early Cash	-2,229***	-2,574***	-1,960	-2,204
	(858)	(885)	(1,719)	(1,763)
T2: Fee-Day Cash	-2,128**	-2,528***	-2,672	-2,348
	(856)	(879)	(1,626)	(1,643)
T3: Voucher	4,019***	3,762***	3,247**	3,659**
	(810)	(838)	(1,653)	(1,681)
Control Mean	5,823	6,223	6,178	6,346
Observations	1,960	1,855	584	564

Notes: Endline re-elicitation of the earmarking premium (voucher valuation − 35,000; grid ±17,500). OLS with strata FE and the baseline premium as control; robust SEs in parentheses. Choice group excluded.

back